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Department of Mathematics and Computer Science
Digital Transformation and Artificial Intelligence 2

Implementation of a Private Cloud using OpenStack Report

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Introduction

OpenStack is a leading open-source platform for building private clouds, offering organizations the flexibility to manage compute, storage, and networking resources. By enabling businesses to create customizable and secure cloud environments, OpenStack has become a preferred solution for organizations that prioritize data control, regulatory compliance, and operational efficiency over the limitations of public cloud offerings.

The deployment of OpenStack follows a systematic approach. It begins with preparing the physical or virtual infrastructure and installing a compatible Linux operating system. Networking is configured to enable seamless communication between nodes(can be servers, each server contains a stack principal components), followed by the installation of OpenStack's core components, including Keystone for identity management, Nova for compute resources, Neutron for networking, and Cinder for storage. Setting up the controller node, deploying compute nodes, and integrating storage and networking are crucial to ensuring a functional environment.

Once the deployment is complete, testing is conducted to verify system functionality, while tools like the Horizon dashboard provide a user-friendly interface for cloud management. Additional security measures, such as firewalls and access controls, combined with monitoring and maintenance tools, ensure the long-term stability and security of the private cloud.

This report outlines the key steps involved in deploying a private OpenStack cloud, providing insights into best practices and challenges to help organizations successfully leverage this powerful platform.

I. Cloud

1. Introduction to Cloud Computing

Cloud computing has revolutionized the way organizations store, access, and manage data and applications. It refers to the delivery of computing resources—such as servers, storage, databases, networking, software, and analytics—over the internet, eliminating the need for on-premises infrastructure. This technology enables on-demand access, scalability, flexibility, and cost-efficiency, making it a cornerstone of modern IT operations.

2. History of Cloud Computing

The concept of cloud computing originated in the 1960s, with the idea of utility computing envisioned by computer scientists like John McCarthy. However, its modern form began to take shape in the late 1990s with the advent of virtualization technology and large-scale internet connectivity. Companies like Salesforce, Amazon Web Services (AWS), and Google pioneered the commercial adoption of cloud services in the early 2000s. Since then, cloud computing has become an essential part of digital transformation worldwide.

3. Types of Cloud Computing

Cloud deployments can be categorized into three main types:

- **Private Cloud**: A private cloud is dedicated to a single organization, offering enhanced security, control, and customization. It is hosted either on-premises or in a privately managed data center. This type of cloud is ideal for organizations with strict compliance and data sovereignty requirements.

  Example: Banks or government institutions using private clouds to ensure sensitive data is protected.

- **Public Cloud**: A public cloud is owned and operated by third-party providers like AWS, Microsoft Azure, or Google Cloud, offering shared resources to multiple users over the internet. Public clouds are scalable and cost-efficient, suitable for businesses seeking to reduce capital expenditure.

  Example: Startups using public clouds to quickly deploy applications without upfront infrastructure costs.

- **Hybrid Cloud**: A hybrid cloud combines private and public cloud environments, enabling data and application sharing between them. This approach offers the flexibility to scale resources based on demand while maintaining critical workloads in a secure private cloud.

  Example: Retail companies managing customer-sensitive data in a private cloud and hosting their e-commerce websites on a public cloud

  .

4. Cloud Computing Services

Cloud computing services are broadly categorized into three main models, each catering to different business needs:

- **Infrastructure as a Service (IaaS)**:
  IaaS provides virtualized computing resources, such as servers, storage, and networking, over the internet. It allows businesses to scale infrastructure on demand without owning physical hardware.

  Examples: AWS EC2, Google Compute Engine, Microsoft Azure Virtual Machines.

  Use Case: Hosting applications, disaster recovery, or running virtual machines.

- **Platform as a Service (PaaS)**:
  PaaS offers a platform for developers to build, test, and deploy applications without managing the underlying infrastructure. It simplifies the development process and reduces the time-to-market.

  Examples: Google App Engine, AWS Elastic Beanstalk, Microsoft Azure App Service.

  Use Case: Building web and mobile applications.

- **Software as a Service (SaaS)**:
  SaaS delivers software applications over the internet on a subscription basis. Users can access these applications via a web browser without worrying about installation, maintenance, or updates.

  Examples: Google Workspace (formerly G Suite), Salesforce, Microsoft 365.

  Use Case: Email, customer relationship management (CRM), and collaboration tools.

5. Conclusion

Cloud computing has become a cornerstone of modern IT strategies, offering diverse deployment models and services to meet varying organizational needs. Whether through private clouds for enhanced security, public clouds for scalability, or hybrid clouds for flexibility, businesses can harness the power of the cloud to achieve operational efficiency and innovation. Understanding the various cloud services—such as IaaS, PaaS, and SaaS—enables organizations to choose solutions that align with their strategic goals, ensuring long-term success in a digital-first world.

II. OpenStack: A Modular Framework for IaaS

OpenStack is an open-source platform designed to deliver Infrastructure as a Service (IaaS), enabling organizations to deploy and manage cloud infrastructure efficiently. It comprises a collection of modular services that work together seamlessly to provide compute, storage, and networking resources. These services are highly flexible and can be deployed individually or in combination, depending on the organization's specific requirements. Below is an overview of its core components:

- ✧ **Keystone (Identity Service)**
  Keystone acts as the central authentication and authorization service in OpenStack. It provides identity management, token-based access control, service cataloging, and policy enforcement. This ensures that only authenticated users and applications can interact with OpenStack resources, enhancing security.

  Example: Managing user access to virtual machines or storage resources.

- ✧ **Nova (Compute Service)**
  Nova is the compute engine of OpenStack, responsible for provisioning and managing virtual machines (VMs) and compute resources. It supports multiple hypervisors, such as KVM, VMware, and Hyper-V, making it versatile for various infrastructure setups.

  Example: Launching virtual servers for hosting applications or workloads.

✧ **Neutron (Networking Service)**
Neutron provides "Networking as a Service" (NaaS) capabilities, allowing users to create and manage complex virtual networks. It supports VLANs, software-defined networking (SDN), and network function virtualization (NFV), enabling advanced network configurations like load balancing and firewalls.

Example: Setting up isolated networks for multi-tenant environments.

✧ **Glance (Image Service)**
Glance handles the storage, discovery, and management of disk images. These images can be used to launch virtual machines, ensuring consistency and rapid deployment. It supports various formats such as RAW, VHD, VMDK, and QCOW2.

Example: Hosting pre-configured images for operating systems and application stacks.

✧ **Cinder (Block Storage Service)**
Cinder provides persistent block storage for virtual machines. It enables the creation, attachment, and management of block storage volumes, which can be resized and retained even after VM termination.

Example: Providing additional storage for database applications or file systems.

✧ **Swift (Object Storage Service)**
Swift offers scalable and redundant object storage for unstructured data such as backups, archives, or media files. It ensures data durability by replicating objects across multiple nodes and data centers.

Example: Storing large datasets, videos, or application logs.

✧ **Horizon (Dashboard)**
Horizon is OpenStack's web-based user interface that simplifies cloud management. It allows users to manage services like virtual machines, networks, and storage through an intuitive dashboard without relying on command-line interfaces.

Example: Creating and managing instances, networks, and storage volumes via a browser.

- ✧ **Customizability and Adaptability**
  OpenStack's modular design makes it highly customizable, allowing organizations to tailor deployments to meet specific needs, whether for small-scale private clouds or large-scale multi-tenant public clouds. By combining these components, businesses can create robust and scalable cloud environments that align with their operational goals.

1. Opensatck Vs AWS VS openNebula

Key Aspect	OpenStack	AWS (Amazon Web Services)	OpenNebula
Type	Open-source cloud platform for IaaS	Public cloud service provider	Open-source cloud platform for IaaS
Deployment Model	Private cloud, public cloud, or hybrid cloud	Public cloud (AWS services), supports hybrid via Outposts	Private cloud, public cloud, or hybrid cloud
Ownership	Community-driven, open-source	Proprietary, operated by Amazon	Community-driven, open-source
Ease of Setup	Complex setup requiring technical expertise	Easy to start with pre-built services	Easier to set up than OpenStack but still requires configuration
Target Users	Enterprises, service providers, developers with control needs	Businesses of all sizes needing scalable, ready-to-use cloud	Small to medium businesses, research institutions
Scalability	Highly scalable but depends on infrastructure	Extremely scalable with global data centers	Scalable but primarily suited for smaller deployments
Cost	Free to use (open-source), with operational costs for hardware	Pay-as-you-go pricing for resources and services	Free to use (open-source), with operational costs for hardware
Ecosystem and Integration	Large ecosystem with support for third-party tools	Comprehensive ecosystem with integrated services	Smaller ecosystem compared to OpenStack and AWS

Key Aspect	OpenStack	AWS (Amazon Web Services)	OpenNebula
Compute Services	Nova for VM management and compute provisioning	Amazon EC2, Lambda for serverless computing	Supports VM and container management via integrated components
Storage Options	Cinder (block storage), Swift (object storage)	S3 (object storage), EBS (block storage), Glacier (archival)	Block, file, and object storage, but more limited than OpenStack
Networking	Neutron for software-defined networking and VLAN management	Amazon VPC for virtual networks	Native support for VLANs and simplified networking
Orchestration	Heat for resource orchestration using templates	AWS CloudFormation for infrastructure templates	Native orchestration with basic features
Monitoring	Ceilometer for metering and monitoring	AWS CloudWatch for detailed monitoring and alerts	Basic monitoring tools, less advanced than OpenStack or AWS
Security	Keystone for identity and access management	Integrated IAM (Identity and Access Management)	Basic identity management, less feature-rich than others
Support and Documentation	Extensive community support, with commercial options available	Professional, paid support with extensive documentation	Smaller community but offers commercial support options
Global Reach	Relies on private infrastructure or hosting provider locations	Global presence with multiple regions and availability zones	Relies on private infrastructure or hosting provider locations
Use Cases	Enterprises needing complete control over their cloud environment	Businesses requiring scalable and ready-to-use cloud services	Organizations seeking a lightweight and cost-effective cloud solution

OpenStack: Best for organizations requiring a highly customizable private cloud solution with complete control over infrastructure. Ideal for large-scale deployments.

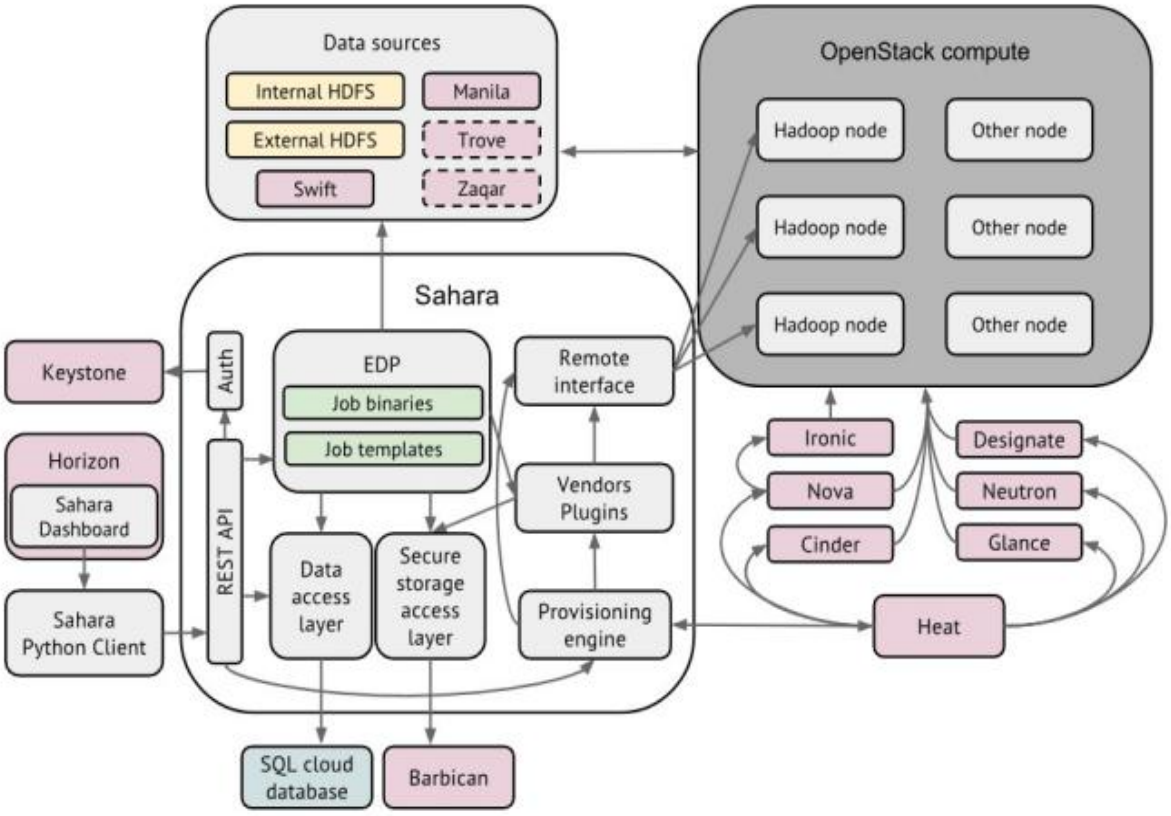
AWS: Suited for businesses seeking an easy-to-use, globally available public cloud with comprehensive services. Requires no upfront investment in infrastructure.

OpenNebula: A simpler alternative to OpenStack, focused on small to medium businesses needing lightweight private cloud solutions.

AWS, **OpenStack**, and **OpenNebula** are prominent cloud computing platforms, each with distinct use cases. **AWS** excels in scalability, reliability, and a broad range of services, making it ideal for enterprises requiring extensive resources, global reach, and advanced tools for artificial intelligence and machine learning. **OpenStack** is an open-source platform designed for organizations seeking customizable private or hybrid cloud solutions, with a focus on flexibility and control. **OpenNebula**, another open-source platform, is simpler to deploy and manage, catering to smaller-scale private clouds or edge computing use cases. Choosing the right platform depends on factors such as the organization's scale, budget, technical expertise, and specific goals.

For a digital transformation and artificial intelligence engineer, AWS is often the best choice due to its advanced AI services and robust ecosystem, though OpenStack may be preferred if the focus is on building highly tailored cloud environments.

III. Architicture OpenStack



This diagram illustrates the integration of various OpenStack components working cohesively to manage and deploy big data processing frameworks, such as Hadoop, using Sahara (the OpenStack Data Processing service). At its core, Sahara serves as a provisioning and management layer, enabling the deployment of data processing clusters on OpenStack compute resources.

It interacts with Keystone for authentication, Horizon for dashboard integration, and Heat for orchestration. Data sources such as Swift (object storage), Manila (file storage),

and HDFS (both internal and external) are utilized for data input and storage. Additional services, including Nova for compute resource management, Cinder for block storage, Neutron for networking, and Glance for image management, play integral roles in provisioning and supporting the infrastructure. Barbican ensures secure storage of sensitive information, while Sahara's plugins and job templates streamline cluster configuration.

Together, these components demonstrate OpenStack's modular architecture and its ability to integrate seamlessly for specialized tasks like big data processing.

Sahara

Role: Sahara is OpenStack's Data Processing service, designed to provision and manage big data frameworks like Hadoop, Spark, and Storm.

How It Works:

Sahara allows users to create and manage data processing clusters with minimal effort.

It integrates with other OpenStack components to orchestrate and provision clusters dynamically.

Key features include job templates, job binaries (scripts, programs), and plugins for various big data engines.

It uses the REST API for communication and enables management via the Sahara Dashboard within Horizon or through the Sahara Python Client.

Keystone

Role: Keystone provides identity, token, catalog, and policy services for OpenStack authentication and authorization.

How It Works:

Keystone ensures secure access by validating user credentials and providing authentication tokens.

Sahara uses Keystone for managing user access and secure integration with other OpenStack components.

Horizon (Sahara Dashboard)

Role: {key Manager switch}Horizon is the web-based dashboard interface for OpenStack, with Sahara integrated for managing data processing tasks.

How It Works:

It provides a graphical user interface for users to configure, deploy, and monitor Sahara clusters.

This reduces the need for complex command-line interactions, making big data provisioning accessible.

Heat

Role: Heat is OpenStack's orchestration service, which automates infrastructure deployment through templates.

How It Works:

Heat coordinates the creation of the virtual machines, networking, and storage resources required for Sahara clusters.

Sahara leverages Heat templates to define and deploy big data clusters efficiently.

OpenStack Compute

Role: Manages compute resources using Nova for provisioning and maintaining virtual machines.

How It Works:

Sahara provisions clusters by launching VMs (nodes) using Nova.

The compute nodes are configured to run as either Hadoop nodes or other roles (e.g., Spark master/slave).

Data Sources

Swift:

Provides object storage for storing unstructured data like logs, input files, or outputs from big data jobs.

Sahara integrates with Swift to use it as a data repository for processing tasks.

Manila:

Offers shared file system storage, enabling data sharing across multiple nodes.

Ideal for cases where applications in the cluster require shared access to data files.

External/Internal HDFS:

Hadoop Distributed File System (HDFS) acts as the primary storage layer for Hadoop clusters.

Sahara can work with both external HDFS (pre-existing setups) or configure internal HDFS storage.

Trove and Zaqar:

Trove provides database as a service (DBaaS), while Zaqar offers messaging as a service for coordination in distributed systems.

Plugins and Job Management

Role: Sahara's Vendors Plugins enable integration with specific big data technologies like Hadoop or Spark.

How It Works:

Plugins provide configurations for deploying clusters tailored to a specific framework.

Job templates define how data processing tasks (like MapReduce jobs) are executed across the cluster.

EDP (Elastic Data Processing) simplifies running data processing workflows.

Cinder (Block Storage)

Role: Provides persistent block storage for instances in the cluster.

How It Works:

Sahara integrates with Cinder to attach block storage volumes to the compute nodes for data-intensive workloads.

Neutron (Networking)

Role: Manages networking for communication between cluster nodes and external systems.

How It Works:

Sahara uses Neutron to configure VLANs, SDN, or flat networks for seamless connectivity between nodes.

Ensures that nodes within a cluster can communicate effectively.

Glance (Image Service)

Role: Manages the VM images used to provision nodes in the Sahara clusters.

How It Works:

Glance stores pre-configured images for Hadoop or other big data frameworks.

Sahara uses these images to quickly deploy clusters without manual configuration of individual nodes.

Barbican (Key Management Service)

Role: Ensures secure storage and management of sensitive information like encryption keys and certificates.

How It Works:

Sahara integrates with Barbican to protect sensitive data and credentials used during cluster provisioning.

Ceph Integration (Not Explicitly Shown)

Role: Offers distributed object, block, and file storage that complements other OpenStack storage solutions like Swift and Cinder.

How It Works:

If integrated, Ceph enhances scalability and redundancy for storage needs in Sahara clusters.

The relationship between these components is as follows:

Sahara uses Nova to provision and manage the VM instances necessary to create data processing clusters.

Data sources can be integrated into the clusters deployed by Sahara to enable the processing of data from various sources.
Nova provides the underlying infrastructure, while Sahara handles the cluster management layer for data processing.

In summary, Nova provides the computing infrastructure in the form of virtual machines, Sahara uses this infrastructure to create data processing clusters, and data sources are processed by these clusters to perform data processing operations.

IV. Create VM

1. Requirements

You need at least :

- RAM: 8 GB
- CPU: with at least 2 cores
- Storage: 80 GB
- OS type to be installed: Linux
- OS version to be installed: Ubuntu-22.04.5

The first step is to install and configure VirtualBox, which is a type 2 hypervisor that allows multiple virtual operating systems (guests) to run on a single host machine.

The tools that will be used throughout this lab are as follows:

- Oracle VM VirtualBox version 7.0.2.
- An ISO file for installing Ubuntu Desktop 22.04.3.

2. Creation and Configuration of a Virtual Machine:

The goal of this first step is to create a new virtual machine using VirtualBox's graphical interface. During this initial step, we will define the basic settings of the virtual machine, including its name, the amount of allocated RAM, and the storage configuration.

- Name: openstack
- RAM: 5 GB
- CPU: 3
- Storage: 80 GB
- OS type to be installed: Linux
- OS version to be installed: Ubuntu-22.04.5

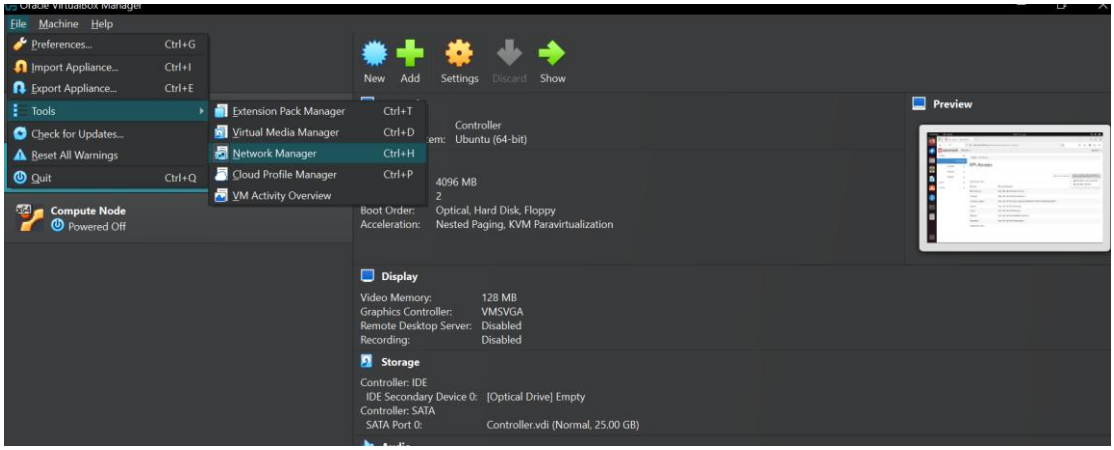
Create a new Vm Machine:

sudo apt update && sudo apt upgrade -y && sudo apt dist-upgrade

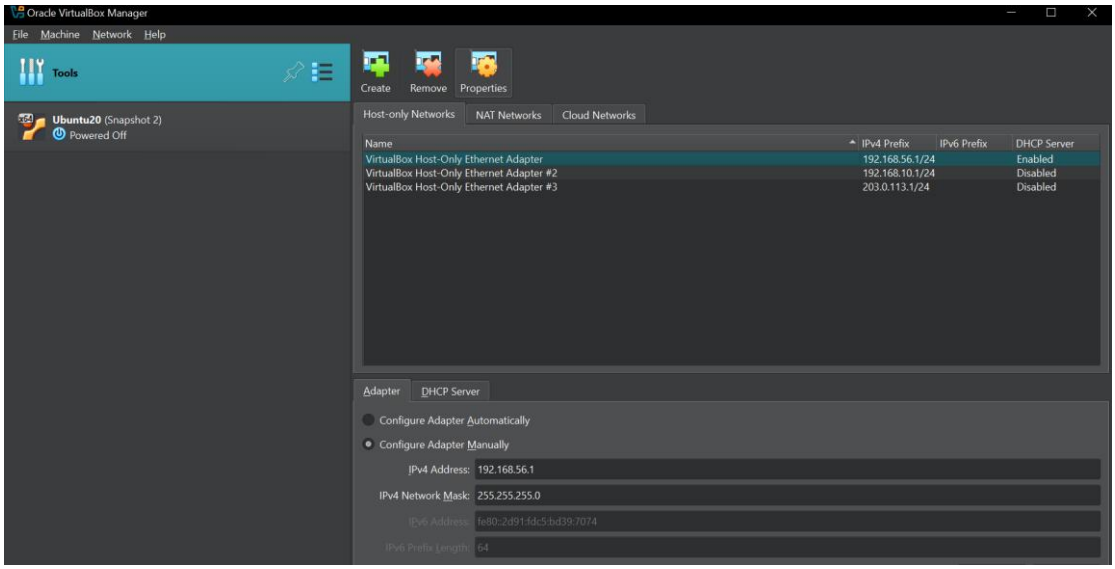
sudo apt-get insatll bridge-utils -y && sudo apt install net-tools -y

3. Configure Network (Multi Adapters [1-Nat, -2host-only])

Access the Network Manager section:

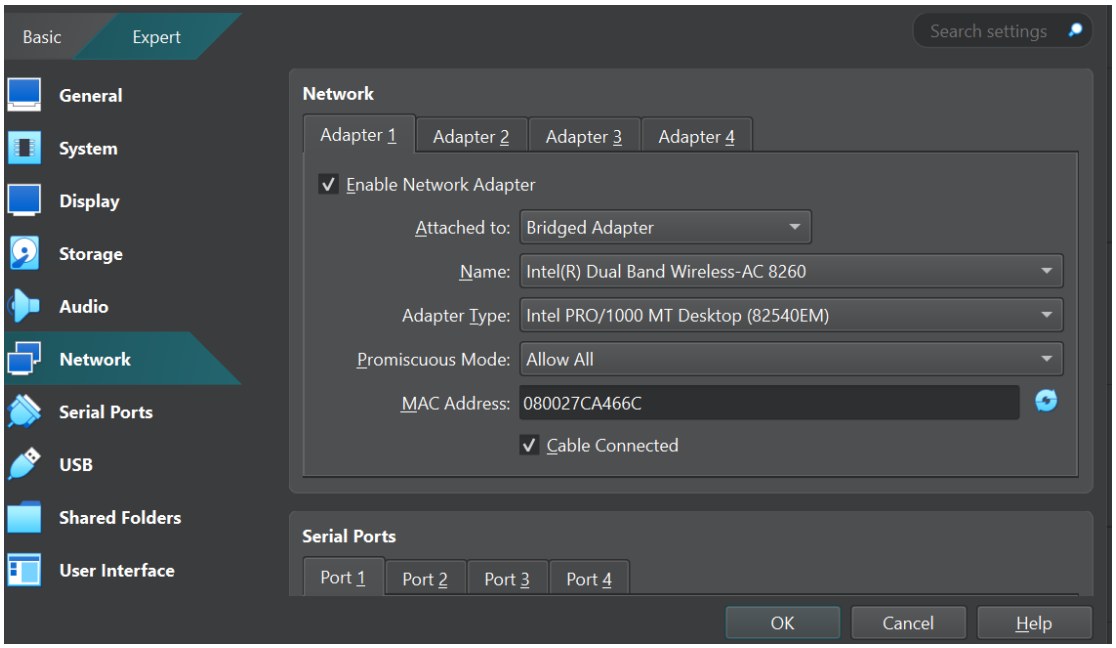


Create a new Network interface and configure it based on your requirements :

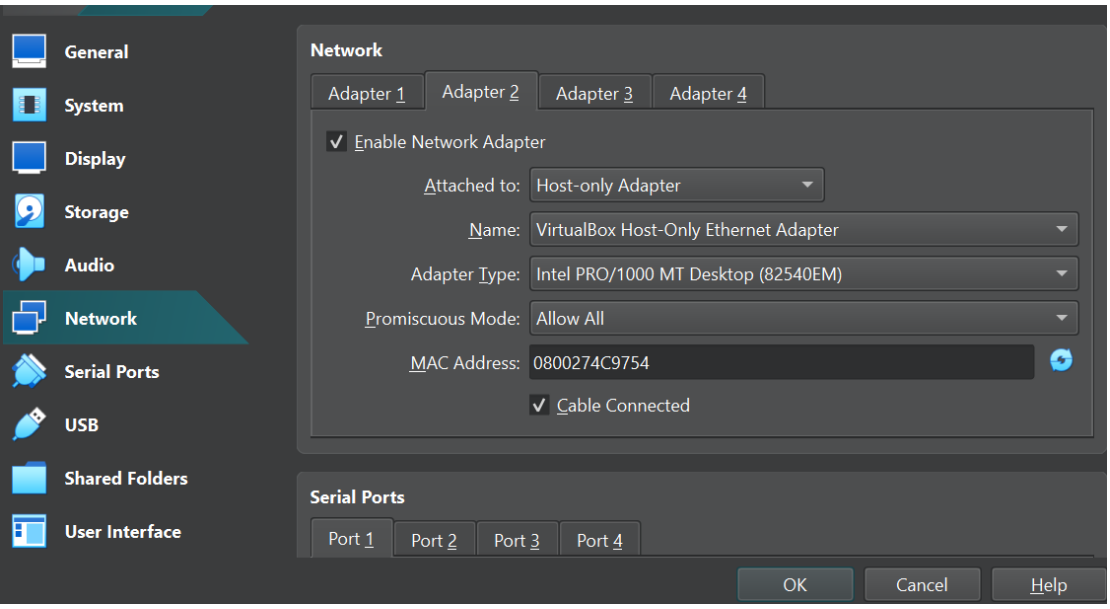


In my case, I created a single network interface ('Virtual Network Adapter') and configured it manually to use the IP address range 192.168.56.1/24.

Ensure that the DHCP is disabled.



First netwrok adapter attached to “Bridge adapter” Mode.



The second adapter is set to 'Host-only Adapter' mode, which will be used by OpenStack’s services to communicate with each other without any interaction with the external network.

Access Your Vm machine and then check the network configurations , using the command “ifconfig” :

```
root@massin-VirtualBox:/home/massin/Downloads# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.2.140  netmask 255.255.255.0  broadcast 192.168.2.255
        inet6 fe80::25ee:e400:decd:9f64  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:ae:f8:9b  txqueuelen 1000  (Ethernet)
        RX packets 6082  bytes 3485030 (3.4 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 2853  bytes 472347 (472.3 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.102  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::27d4:1e6d:e07:75d8  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:c0:ec:eb  txqueuelen 1000  (Ethernet)
        RX packets 130  bytes 25652 (25.6 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 186  bytes 25763 (25.7 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 185626  bytes 44927239 (44.9 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 185626  bytes 44927239 (44.9 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

In this configuration, there are two main network adapters: **enp0s3**, configured with the IP range 192.168.2.x, and **enp0s8**, configured with the IP range 192.168.56.x. Both adapters are active and assigned unique subnets.

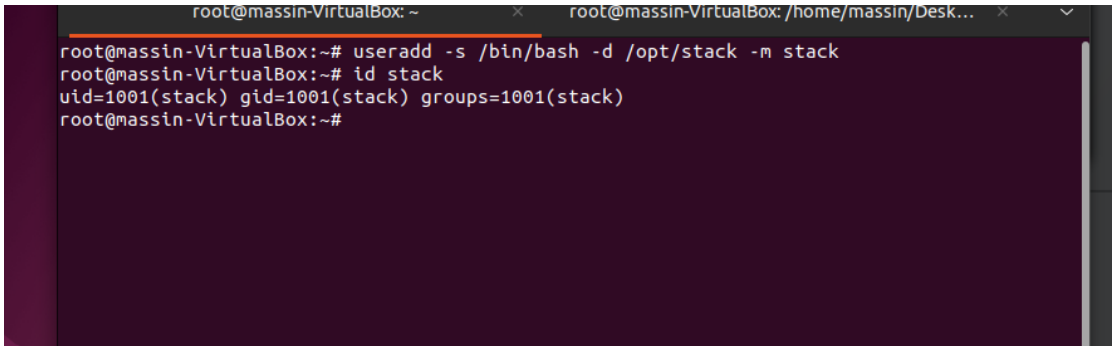
v. Install OpenStack

1. Creating a Non-Root User

DevStack requires a non-root user with sudo privileges.

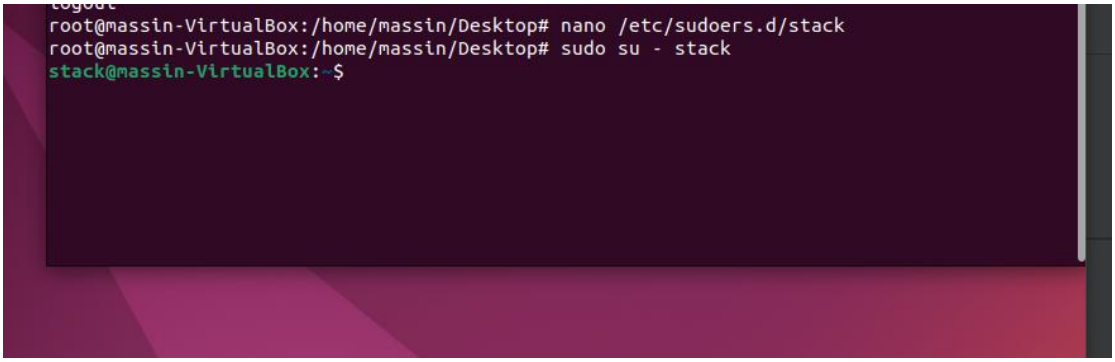
Create a user:

```
sudo useradd -s /bin/bash -d /opt/stack -m stack
echo "stack ALL=(ALL) NOPASSWD:ALL" | sudo tee
/etc/sudoers.d/stack
```



Switch to the `stack` user:

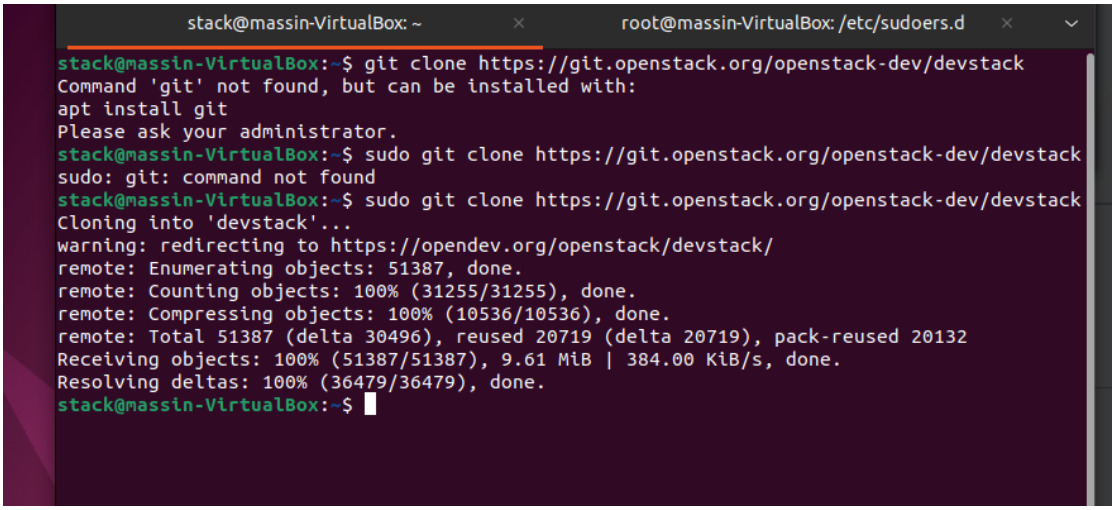
su - stack



2. Cloning the DevStack Repository

Clone the official DevStack repository:

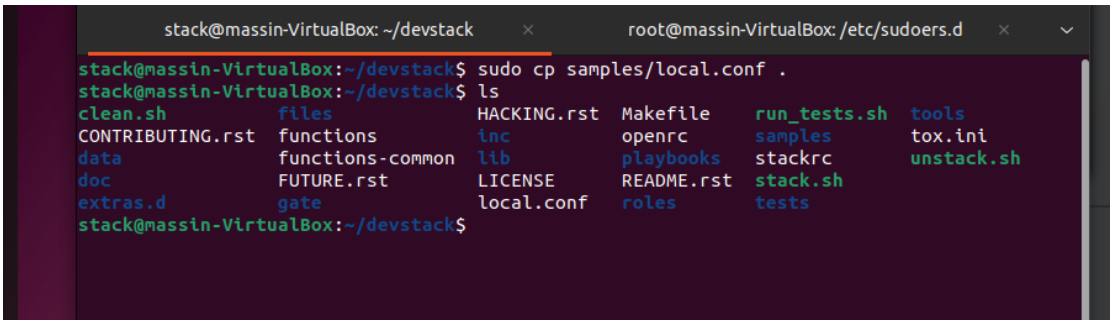
```
git clone https://opendev.org/openstack/devstack

cd devstack
```



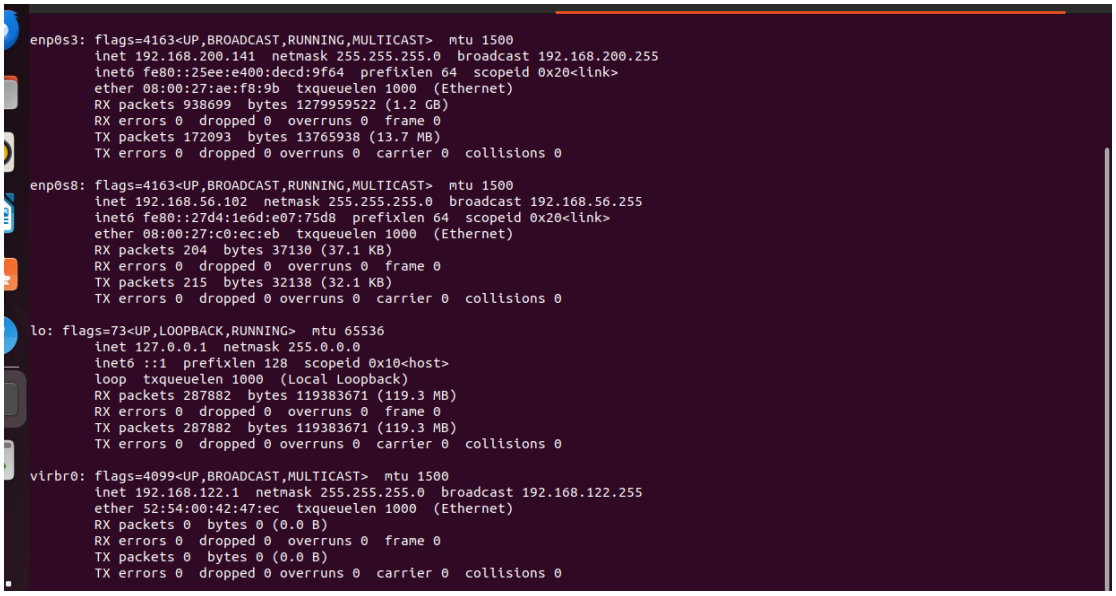
3. Configuring the `` File``

The `local.conf` file is used to customize the DevStack setup.

Copy the file:



Get Address Ip and name of The network interface :



Modify the file **local.conf :**

The local.conf file is a configuration file for DevStack. It allows the user to specify various configuration parameters such as which OpenStack services to enable, IP addresses, passwords, etc. It plays a crucial role in customizing the OpenStack installation via DevStack.

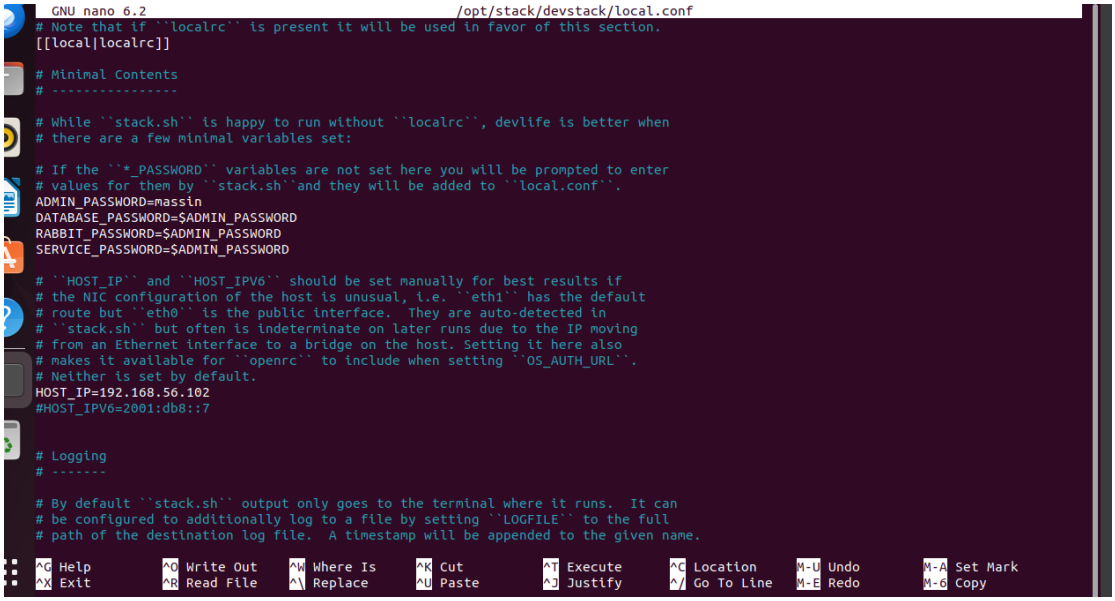
So, we make some modifications to the local.conf file to customize the settings:

=> Set passwords :

These passwords are required for administrative access, database, RabbitMQ, and OpenStack services. Replace “massin” with a secure and unique password in production environments.

=> Set Host ip:

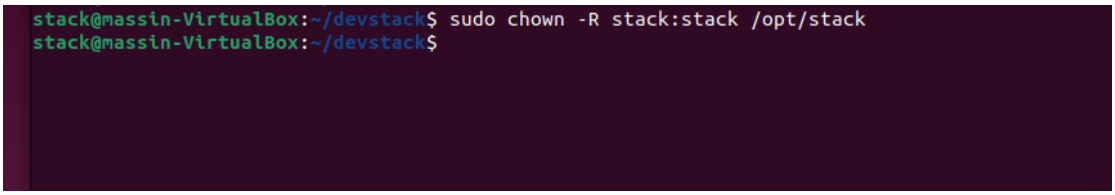
set The <**HOST_IP**> with your system's local IP (e.g., 192.168.56.x)



change the ownership of /opt/devstack

Then, we change the ownership of the /opt/devstack directory and all of its contents to the user 'stack' and the group 'stack'. This is because when deploying OpenStack with DevStack, the installation directory requires specific permissions for the stack user to ensure the proper functioning of the OpenStack services.

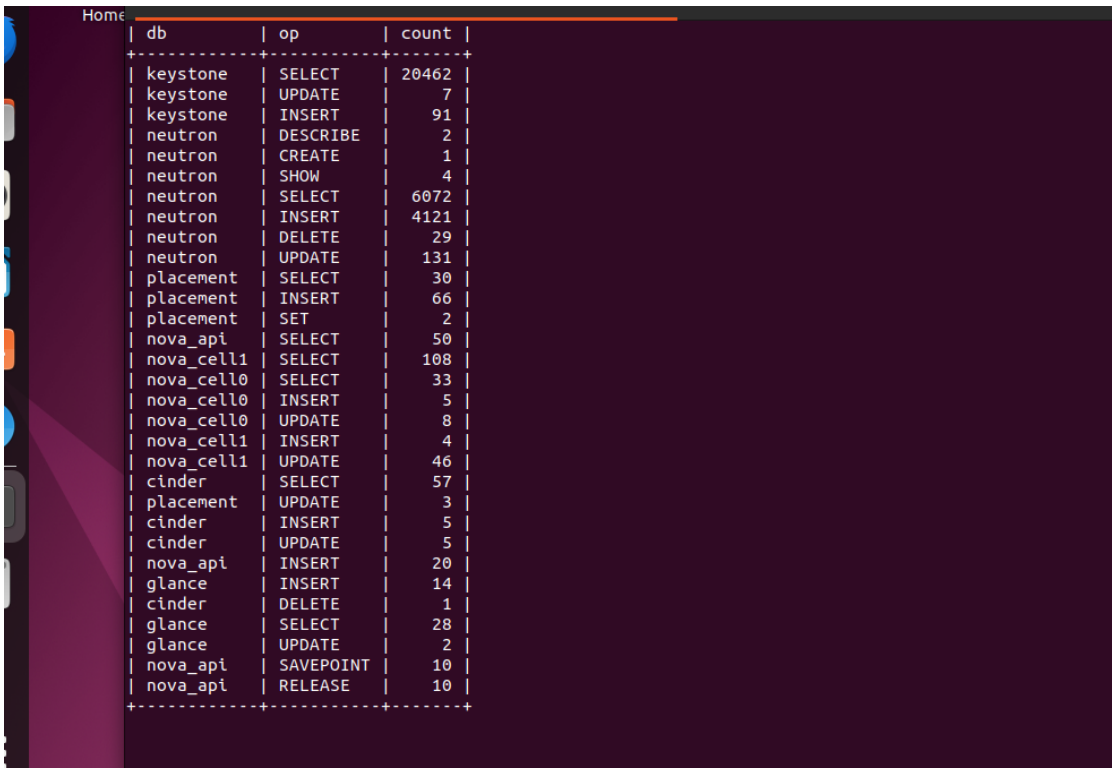
```
$ sudo chown -R stack:stack /opt/stack
```



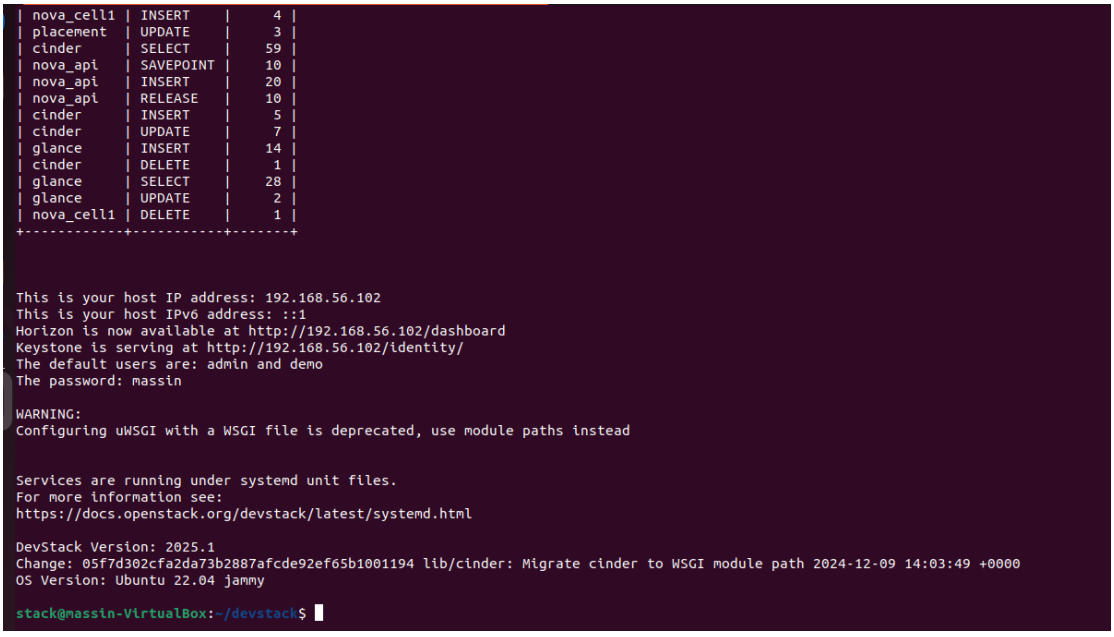
Then, we run the stack.sh script:

```
stack@massin-VirtualBox:~/devstack$ sudo chown -R stack:stack /opt/stack
stack@massin-VirtualBox:~/devstack$ ./stack.sh
+ unset GREP_OPTIONS
+ unset LANG
+ unset LANGUAGE
+ LC_ALL=en_US.utf8
+ export LC_ALL
++ env
++ grep -E '^OS_'
++ cut -d = -f 1
+ unset
+ umask 022
+ PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin:/usr/local/bin:/
/sbin:/sbin
+++ dirname ./stack.sh
++ cd .
++ pwd
+ TOP_DIR=/opt/stack/devstack
+ NOUNSET=
+ [[ -n '' ]]
++ date +%s
+ DEVSTACK_START_TIME=1733885670
+ [[ -r /opt/stack/devstack/.stackenv ]]
+ FILES=/opt/stack/devstack/files
+ '[' '!' -d /opt/stack/devstack/files ']'
+ '[' '!' -d /opt/stack/devstack/inc ']'
+ '[' '!' -d /opt/stack/devstack/lib ']'
+ [[ '' == \y ]]
+ [[ 1001 -eq 0 ]]
+ [[ -n '' ]]
+ [[ -e /opt/stack/.no-devstack ]]
```

```
Unpacking liburing2:amd64 (2.1-2build1) ...
Selecting previously unselected package libusbredirparser1:amd64.
Preparing to unpack .../27-libusbredirparser1_0.11.0-2build1_amd64.deb ...
Unpacking libusbredirparser1:amd64 (0.11.0-2build1) ...
Selecting previously unselected package libvirglrenderer1:amd64.
Preparing to unpack .../28-libvirglrenderer1_0.9.1-1~exp1ubuntu2_amd64.deb ...
Unpacking libvirglrenderer1:amd64 (0.9.1-1~exp1ubuntu2) ...
Selecting previously unselected package libfdt1:amd64.
Preparing to unpack .../29-libfdt1_1.6.1-1_amd64.deb ...
Unpacking libfdt1:amd64 (1.6.1-1) ...
Selecting previously unselected package qemu-system-common.
Preparing to unpack .../30-qemu-system-common_1%3a6.2+dfsg-2ubuntu6.24_amd64.deb ...
Unpacking qemu-system-common (1:6.2+dfsg-2ubuntu6.24) ...
Selecting previously unselected package qemu-system-data.
Preparing to unpack .../31-qemu-system-data_1%3a6.2+dfsg-2ubuntu6.24_all.deb ...
Unpacking qemu-system-data (1:6.2+dfsg-2ubuntu6.24) ...
Selecting previously unselected package seabios.
Preparing to unpack .../32-seabios_1.15.0-1_all.deb ...
Unpacking seabios (1.15.0-1) ...
Selecting previously unselected package qemu-system-x86.
Preparing to unpack .../33-qemu-system-x86_1%3a6.2+dfsg-2ubuntu6.24_amd64.deb ...
Unpacking qemu-system-x86 (1:6.2+dfsg-2ubuntu6.24) ...
Selecting previously unselected package qemu-utils.
Preparing to unpack .../34-qemu-utils_1%3a6.2+dfsg-2ubuntu6.24_amd64.deb ...
Unpacking qemu-utils (1:6.2+dfsg-2ubuntu6.24) ...
Selecting previously unselected package qemu-block-extra.
Preparing to unpack .../35-qemu-block-extra_1%3a6.2+dfsg-2ubuntu6.24_amd64.deb ...
Unpacking qemu-block-extra (1:6.2+dfsg-2ubuntu6.24) ...
Selecting previously unselected package qemu-system-gui.
Preparing to unpack .../36-qemu-system-gui_1%3a6.2+dfsg-2ubuntu6.24_amd64.deb ...
Unpacking qemu-system-gui (1:6.2+dfsg-2ubuntu6.24) ...
Selecting previously unselected package ovmf.
Preparing to unpack .../37-ovmf_2022.02-3ubuntu0.22.04.3_all.deb ...
Unpacking ovmf (2022.02-3ubuntu0.22.04.3) ...
Setting up libibverbs1:amd64 (39.0-1) ...
```



db	op	count
keystone	SELECT	20462
keystone	UPDATE	7
keystone	INSERT	91
neutron	DESCRIBE	2
neutron	CREATE	1
neutron	SHOW	4
neutron	SELECT	6072
neutron	INSERT	4121
neutron	DELETE	29
neutron	UPDATE	131
placement	SELECT	30
placement	INSERT	66
placement	SET	2
nova_api	SELECT	50
nova_cell1	SELECT	108
nova_cell0	SELECT	33
nova_cell0	INSERT	5
nova_cell0	UPDATE	8
nova_cell1	INSERT	4
nova_cell1	UPDATE	46
cinder	SELECT	57
placement	UPDATE	3
cinder	INSERT	5
cinder	UPDATE	5
nova_api	INSERT	20
glance	INSERT	14
cinder	DELETE	1
glance	SELECT	28
glance	UPDATE	2
nova_api	SAVEPOINT	10
nova_api	RELEASE	10



nova_cell1	INSERT	4
placement	UPDATE	3
cinder	SELECT	59
nova_api	SAVEPOINT	10
nova_api	INSERT	20
nova_api	RELEASE	10
cinder	INSERT	5
cinder	UPDATE	7
glance	INSERT	14
cinder	DELETE	1
glance	SELECT	28
glance	UPDATE	2
nova_cell1	DELETE	1

```
This is your host IP address: 192.168.56.102
This is your host IPv6 address: ::1
Horizon is now available at http://192.168.56.102/dashboard
Keystone is serving at http://192.168.56.102/identity/
The default users are: admin and demo
The password: massin

WARNING:
Configuring uWSGI with a WSGI file is deprecated, use module paths instead


Services are running under systemd unit files.
For more information see:
https://docs.openstack.org/devstack/latest/systemd.html

DevStack Version: 2025.1
Change: 05f7d302cfa2da73b2887afcde92ef65b1001194 lib/cinder: Migrate cinder to WSGI module path 2024-12-09 14:03:49 +0000
OS Version: Ubuntu 22.04 jammy

stack@massin-VirtualBox:~/devstack$
```

The **stack.sh** script is the main script used by DevStack to deploy an OpenStack environment on a machine. When you run the stack.sh script, several steps are performed, including:

System Configuration: The script begins by checking and configuring various system settings to ensure that the machine is ready for OpenStack deployment. This may include network configuration, checking software dependencies, etc.

Downloading OpenStack Components: The script downloads the required OpenStack components from the appropriate Git repositories. This includes services such as Nova (compute), Neutron (networking), Glance (image service), Keystone (identity), and others.

Configuring OpenStack Services: Once the components are downloaded, the stack.sh script configures these services according to the parameters specified in the local.conf configuration file. This may include parameters such as IP addresses, passwords, and choices of services to enable, etc.

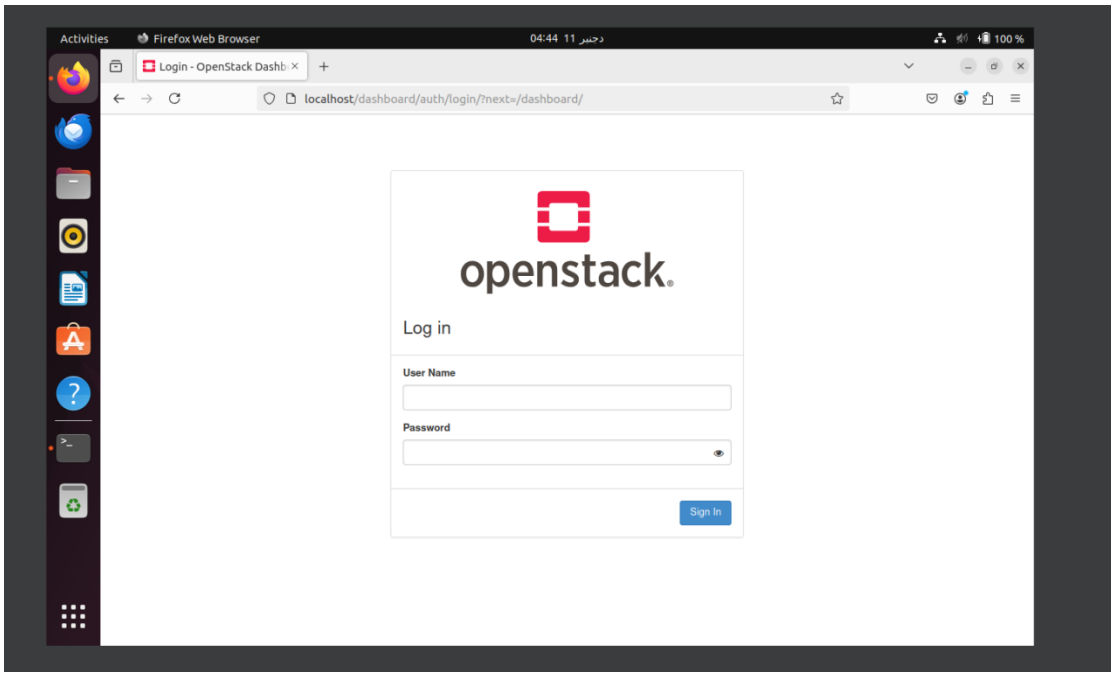
Starting OpenStack Services: Once configured, the OpenStack services are started. This means that the various services, such as Nova, Neutron, and others, are launched as processes on the machine.

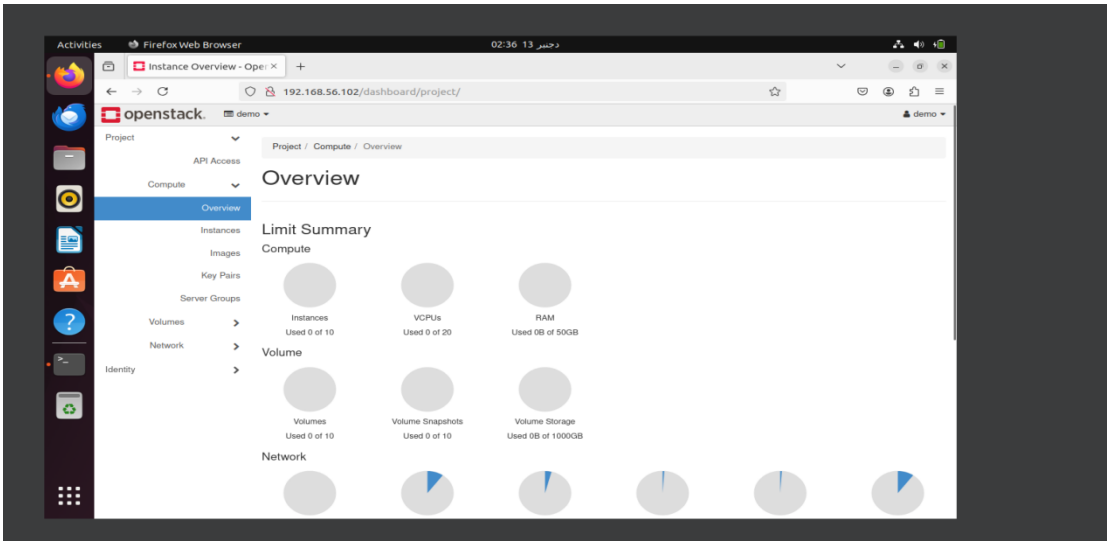
Creating Projects, Users, and Roles: The script typically creates projects, users, and roles in OpenStack to simulate a multi-user environment.

Creating Images: Images of instances (virtual machines) may be downloaded and saved for use by the deployed instances.

Configuring the Client Environment: The script often generates an openrc or stackrc file that can be sourced in the current terminal environment to configure the necessary environment variables to interact with the newly deployed OpenStack environment.

DevStack typically installs the Horizon dashboard, which is a web interface for OpenStack. You can access the dashboard by opening a web browser and going to the IP address of your machine (by default, this is often http://localhost/dashboard).





Next, we need to download the script **`admin-openrc.sh`** from the OpenStack graphical interface so that we can display all the installed services using the following commands:

4. Configures the environment of OpenStack

Then, the **admin-openrc.sh** script must be downloaded from the OpenStack graphical interface so that we can display all the installed services.

The file admin-openrc.sh is OpenStack RC file (resource configuration file) used to

set environment variables required for interacting with OpenStack services via the command line. file typically corresponds to a specific user and project with different roles and permissions.

5. Key Differences Between admin-openrc.sh and demo-openrc.sh

User Role and Permissions:

- `admin-openrc.sh`:
  - Used for administrative tasks.
  - Configures the environment variables for a user with administrative privileges (usually the `admin` user).
  - Allows full access to OpenStack services, including creating projects, users, roles, and managing other resources.
- `demo-openrc.sh`:
  - Used for a regular (non-administrative) user, typically a demo or test user.
  - Configures environment variables for a user with restricted access based on the assigned roles.
  - Limited to managing resources (like instances, volumes, and networks) within a specific project.

Scope of Operations:

- The `admin-openrc.sh` file gives access to all projects and administrative APIs.
- The `demo-openrc.sh` file is scoped to a specific project (e.g., `demo`), and the user is restricted to their project's resources.

Environment Variables:

- Both files set variables like `OS_USERNAME`, `OS_PROJECT_NAME`, `OS_PASSWORD`, `OS_AUTH_URL`, and others.
- The values differ depending on the user and project associated with the file

Example snippet from admin-openrc.sh:

```
export OS_USERNAME=admin
export OS_PROJECT_NAME=admin
export OS_PASSWORD=your_admin_password
export OS_AUTH_URL=http://controller:5000/v3
```

Example snippet from demo-openrc.sh:

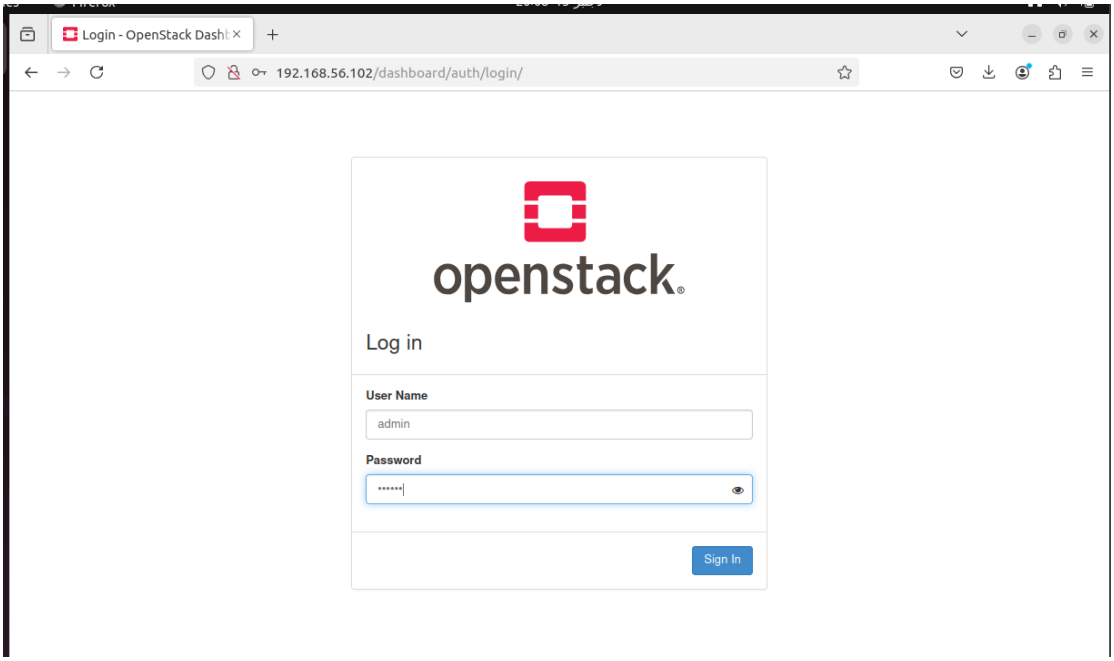
```
export OS_USERNAME=demo
export OS_PROJECT_NAME=demo
export OS_PASSWORD=your_demo_password
export OS_AUTH_URL=http://controller:5000/v3
```

- Use admin-openrc.sh. This is required for tasks like creating new projects, managing users, assigning roles, or setting quotas.

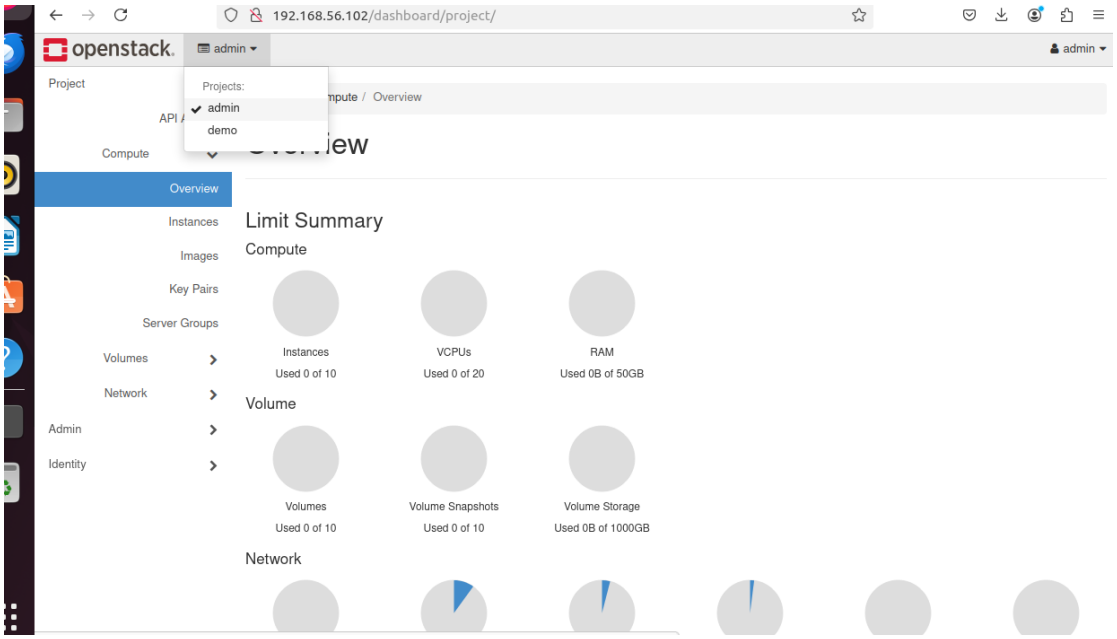
- Use demo-openrc.sh if you're only working within the demo project and don't need administrative-level operations.

Steps :

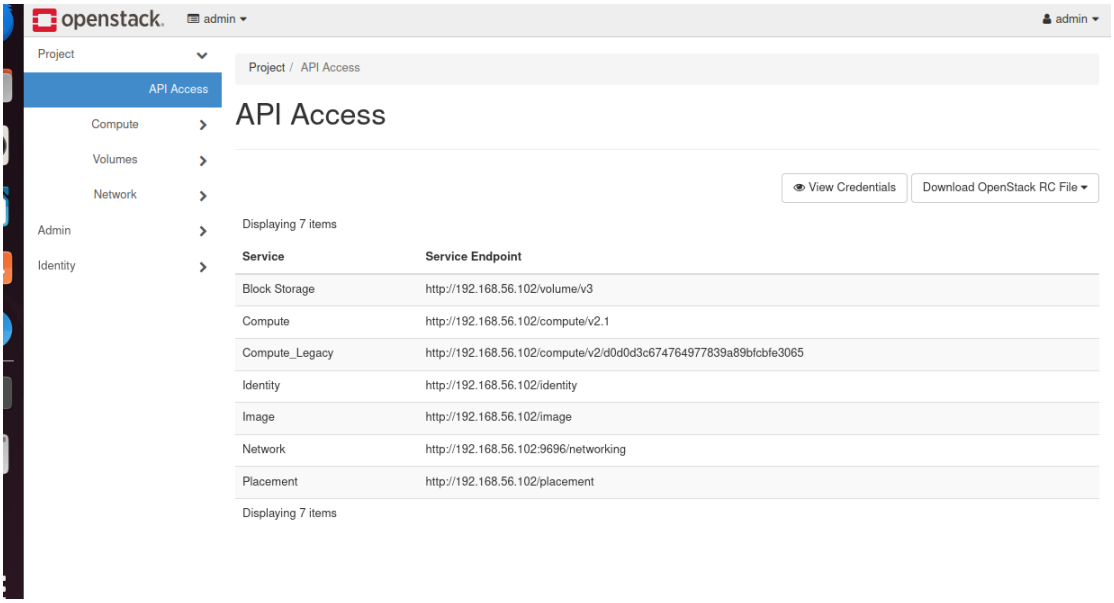
1. Connect to dashboard using admin credentials:



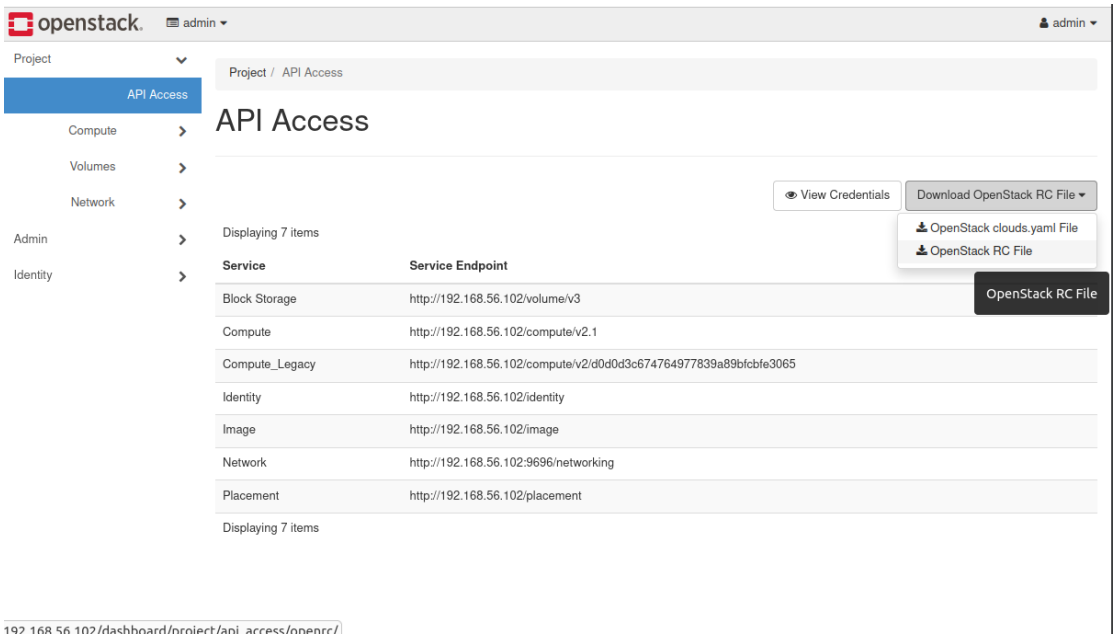
2. Chose admin project:



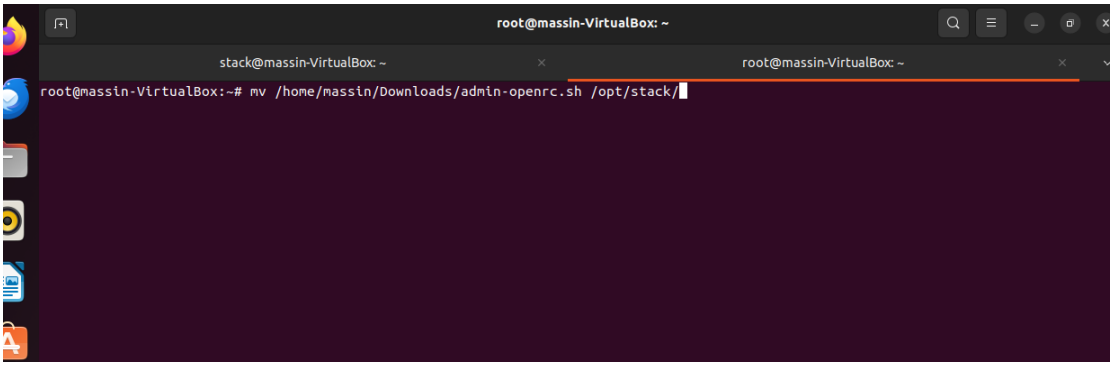
3. Access "**Api Access**" Interface :



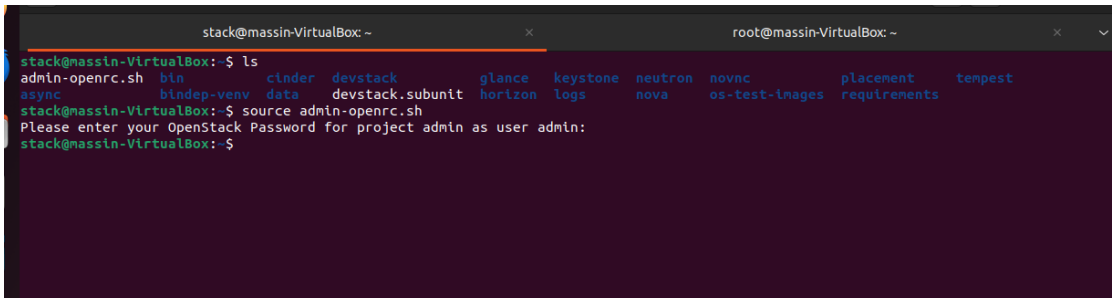
4. Install admin-openrc.sh file:



5. Move the file into stack home directory:

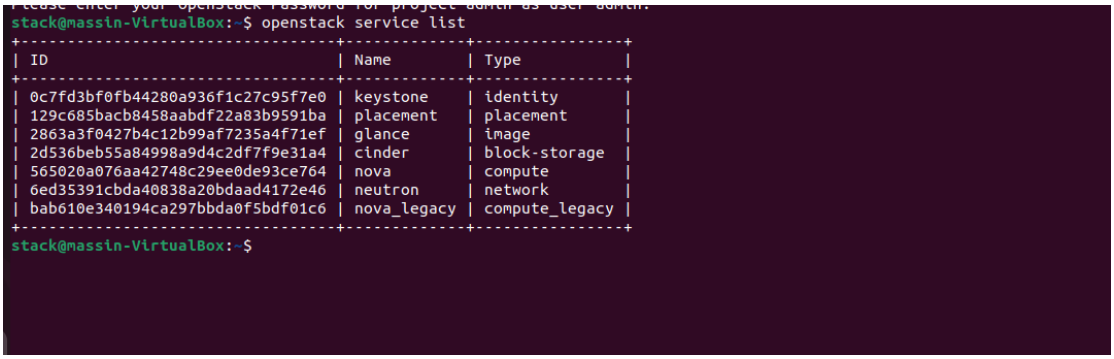


6. Source the file :



Sourcing the RC file (e.g., `admin-openrc.sh` or `demo-openrc.sh`) sets environment variables required for authentication and interaction with OpenStack services via the command-line interface. These variables include details like the username, project name, password, and authentication endpoint, which eliminate the need to specify them manually for each command. By sourcing the RC file, the user gains access to OpenStack resources and permissions based on their role and project scope. For example, the `admin-openrc.sh` provides administrative privileges, while the `demo-openrc.sh` limits access to resources within the `demo` project.

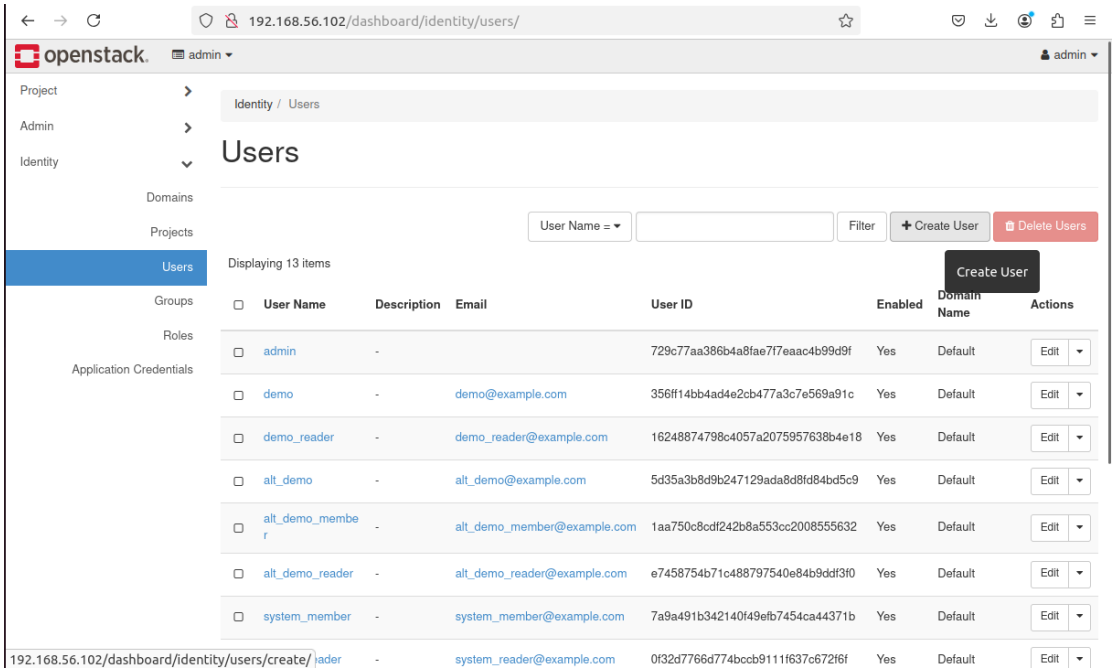
Run Command [openstack service list]



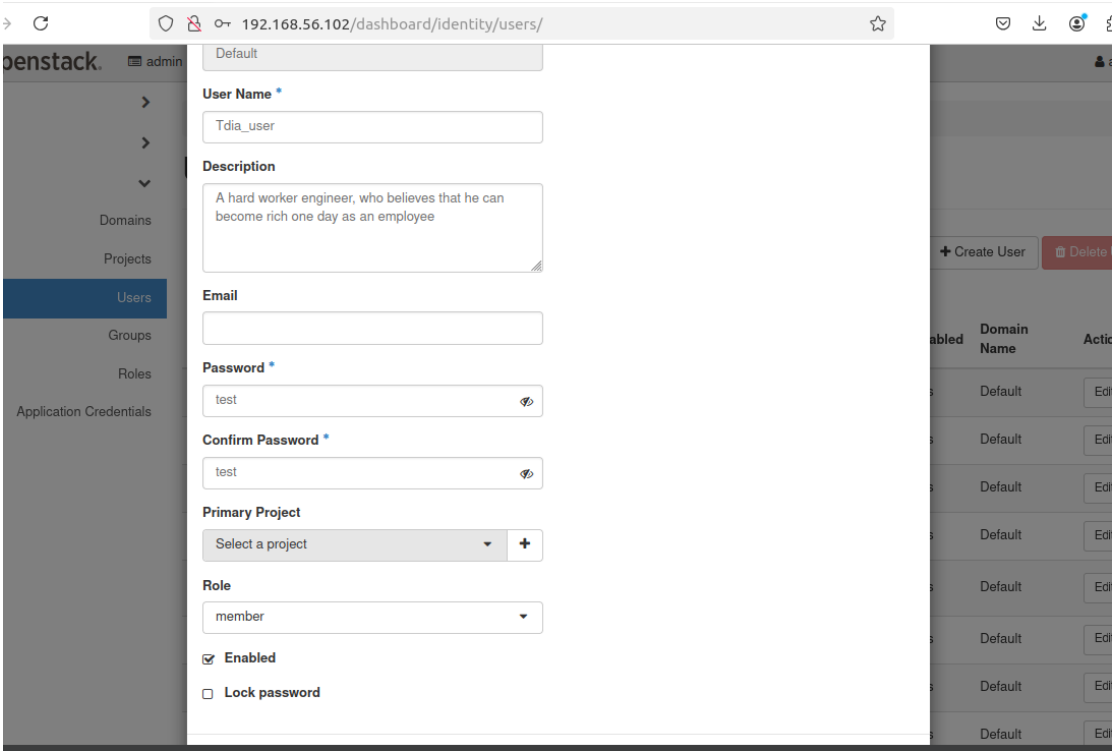
is used to display a list of all services registered in the OpenStack Identity service (Keystone). It provides information about the services that are available in the OpenStack environment and their corresponding API endpoints.

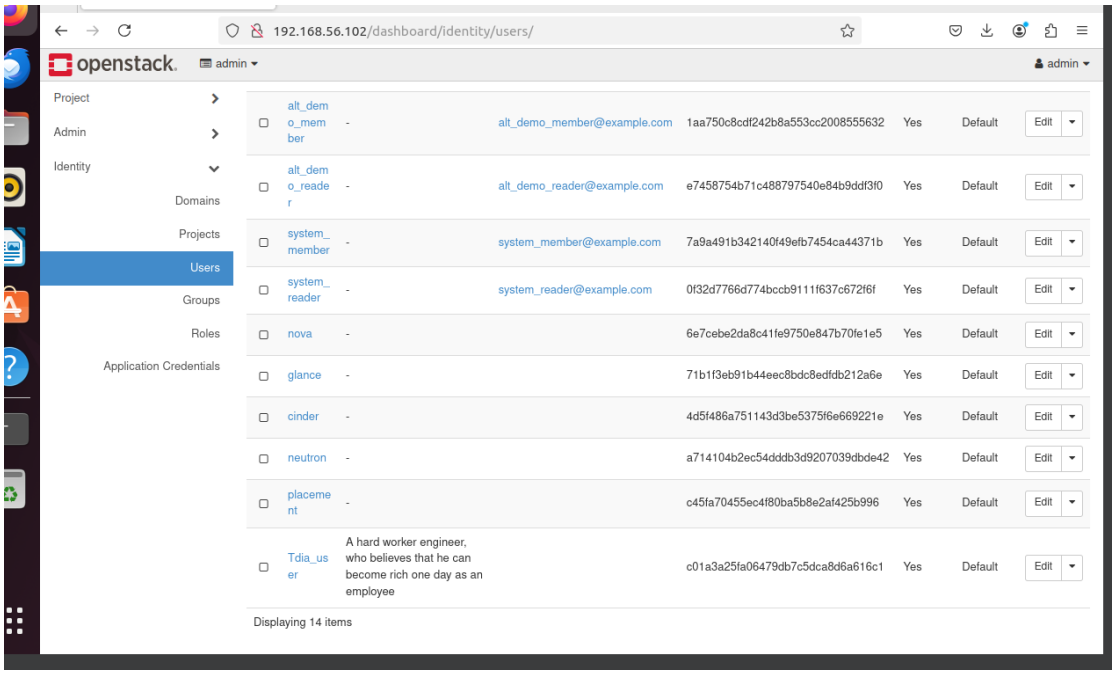
VI. Creation of instances

1. Creation of an Ubuntu instance:



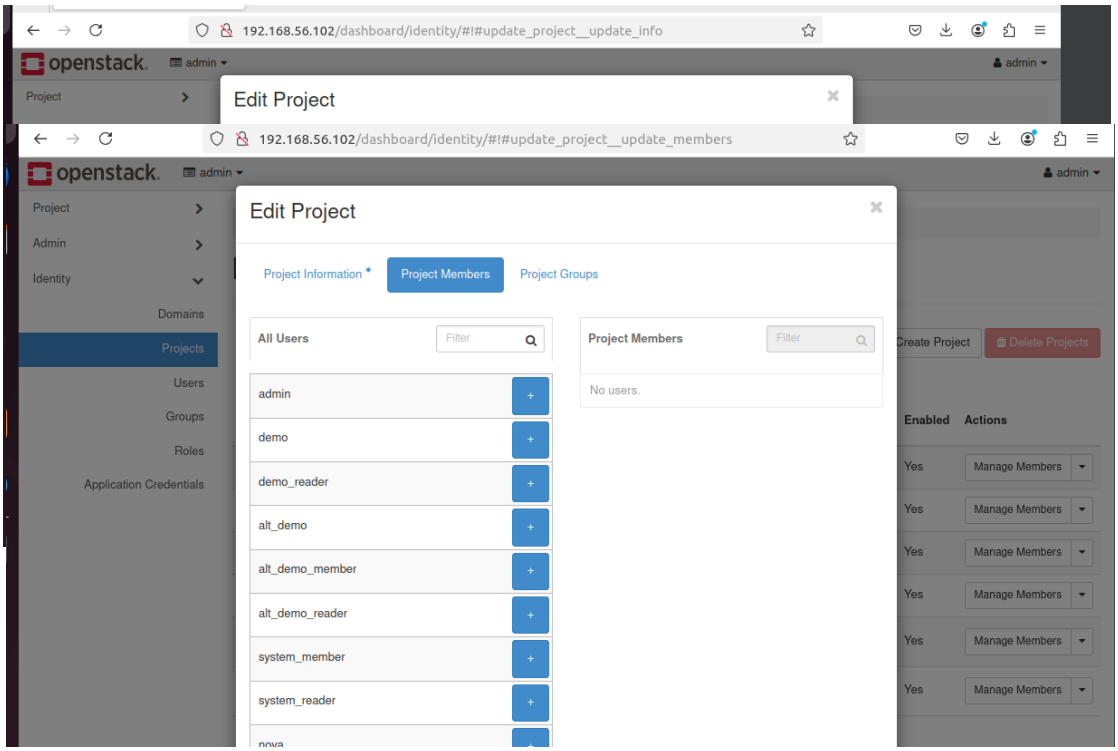
Connect as Admin user and then navigate to “Users” interface under **“identity”** section , and creat a new user .


192.168.56.102/dashboard/identity/users/
openstack.
admin
Default
User Name *
Tdia_user
Description
A hard worker engineer, who believes that he can become rich one day as an employee
Email
Password *
test
Confirm Password *
test
Primary Project
Select a project
Role
member
Enabled
Lock password
Domains
Projects
Users
Groups
Roles
Application Credentials
+ Create User
Domain Name
Default


192.168.56.102/dashboard/identity/users/
openstack.
admin
Project
Admin
Identity
Domains
Projects
Users
Groups
Roles
Application Credentials
alt_demo_member
alt_demo_member@example.com
1aa750c8cdf242b8a553cc2008555632
alt_demo_reader
alt_demo_reader@example.com
e7458754b71c488797540e84b9ddf3f0
system_member
system_member@example.com
7a9a491b342140f49efb7454ca44371b
system_reader
system_reader@example.com
0f32d7766d774bccb9111f637c672f6f
nova
6e7cebe2da8c41fe9750e847b70fe1e5
glance
71b1f3eb91b44eec8bdc8edfdb212a6e
cinder
4d5f486a751143d3be5375f6e669221e
neutron
a714104b2ec54dddb3d9207039dbde42
placement
c45fa70455ec4f80ba5b8e2af425b996
Tdia_user
A hard worker engineer, who believes that he can become rich one day as an employee
c01a3a25fa06479db7c5dca8d6a616c1
Yes
Default
Edit
Displaying 14 items

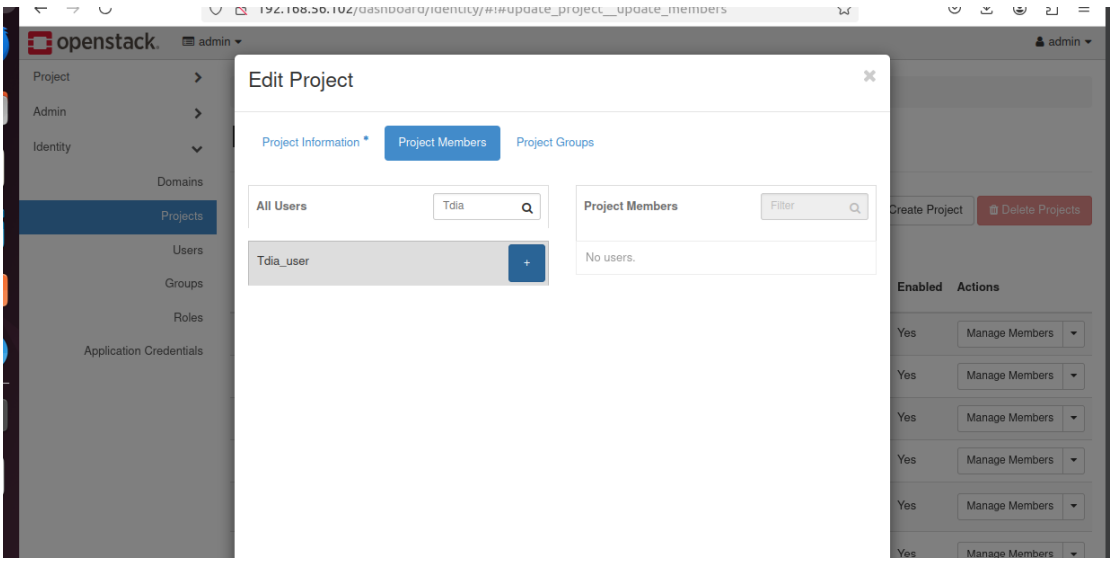
2. Creation of a New Project:

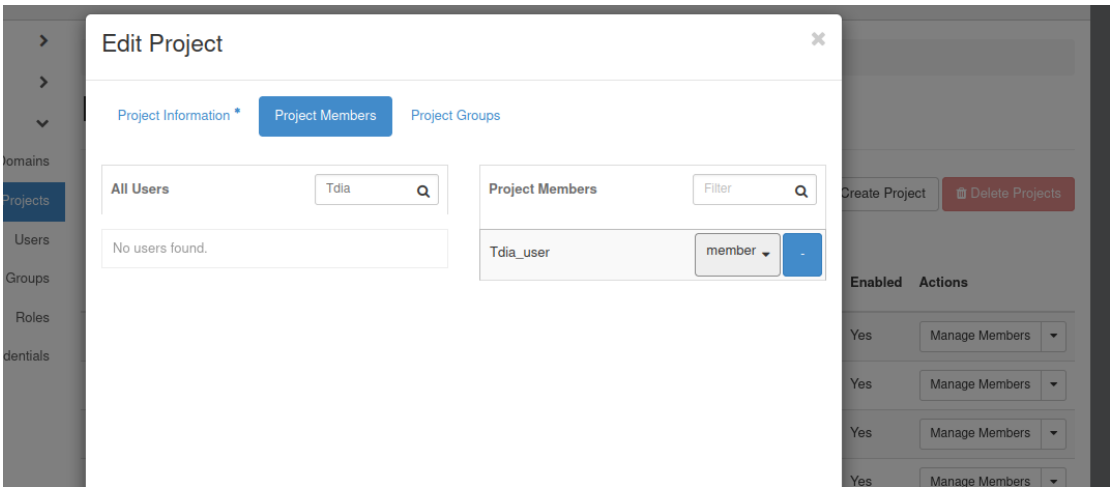
Set a Name and a description(optional) to the project : Tdia_user

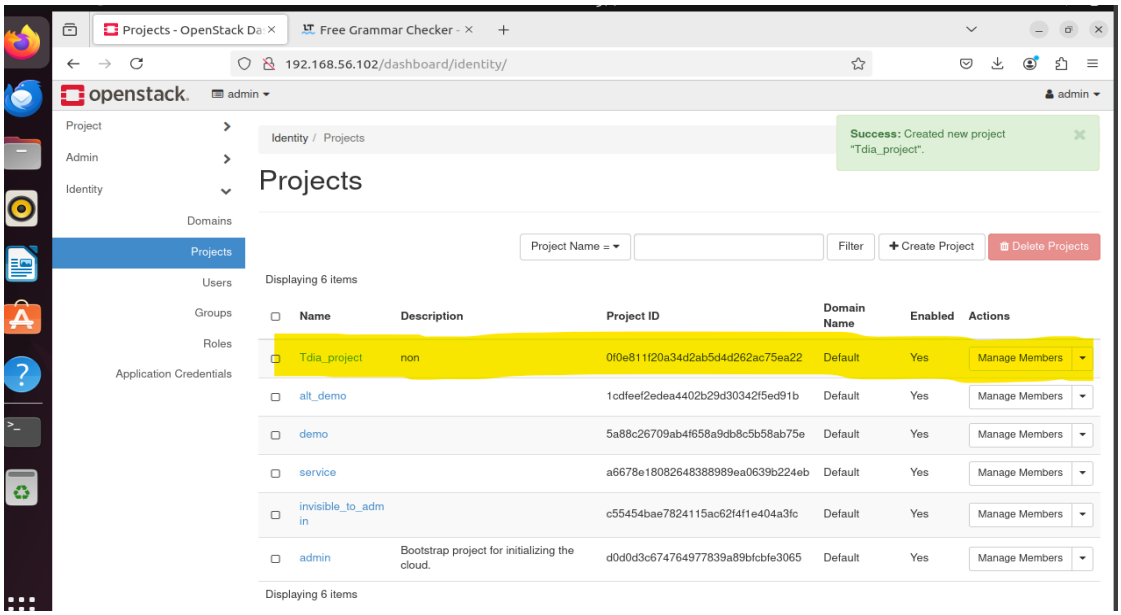


Set Restriction of Access to the Project :(Only our user will have access to the project)

Affect the User created before to the project :

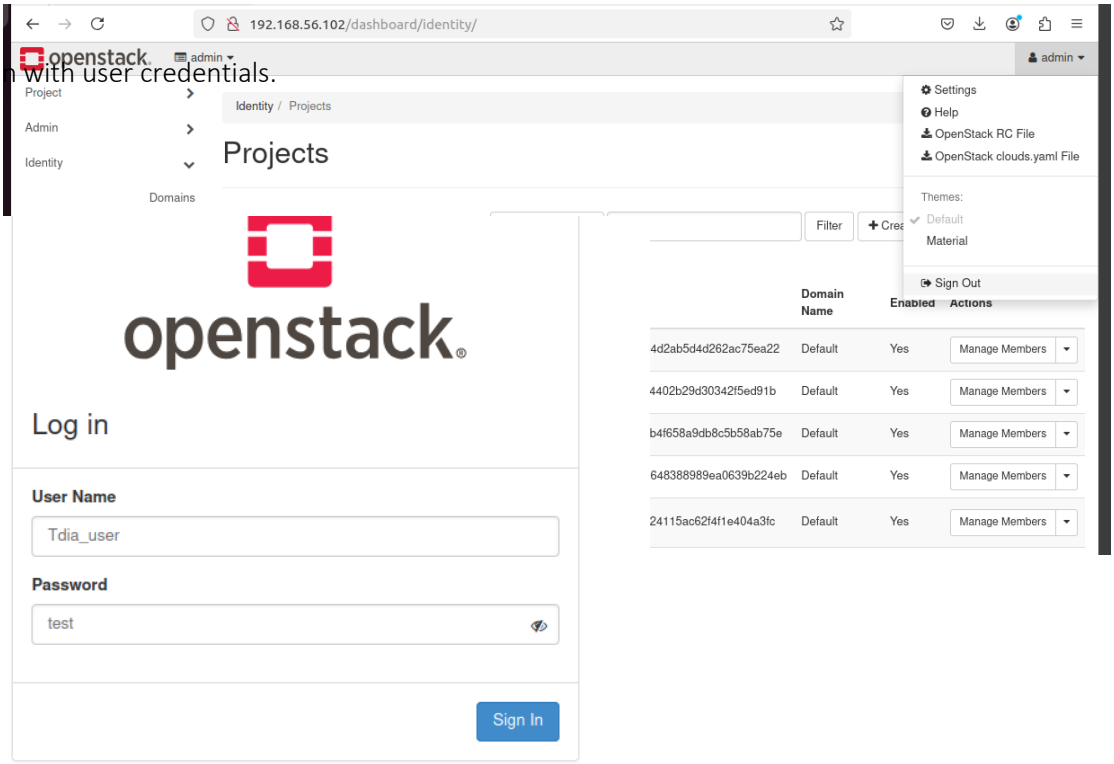


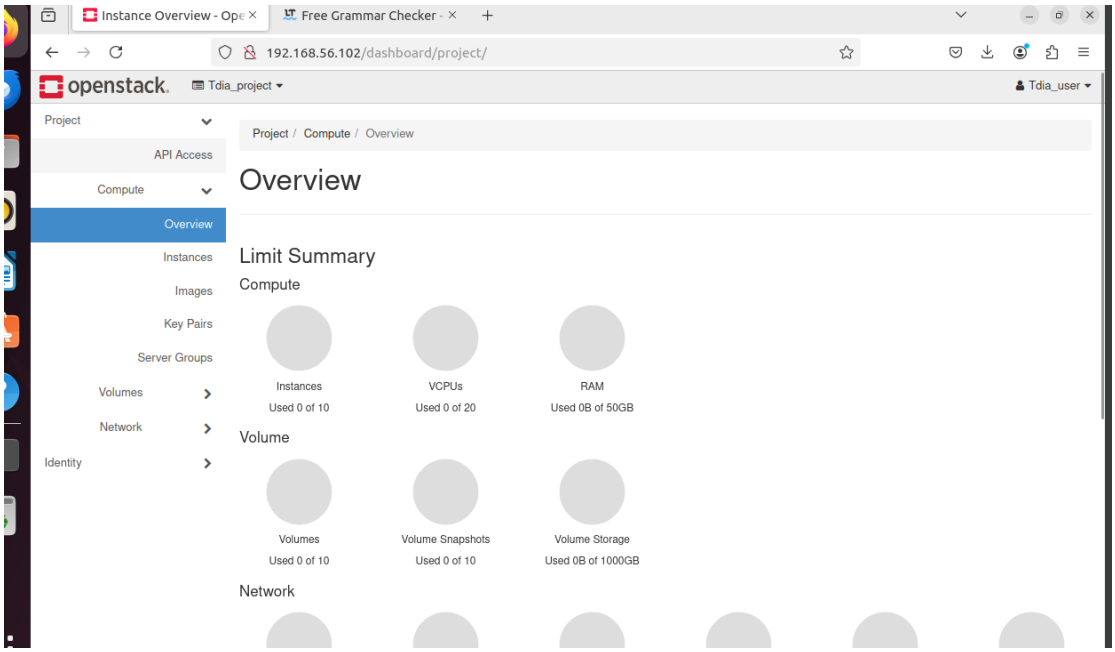

Edit Project
Project Information *
Project Members
Project Groups
All Users
Tdia
No users found.
Project Members
Filter
Tdia_user
member
Create Project
Delete Projects
Enabled
Actions
Yes
Manage Members
Domains
Projects
Users
Groups
Roles
dentials


Projects - OpenStack Da
Free Grammar Checker -
192.168.56.102/dashboard/identity/
openstack.
admin
Project
Admin
Identity
Domains
Projects
Users
Groups
Roles
Application Credentials
Identity / Projects
Success: Created new project "Tdia_project".
Projects
Project Name =
Filter
+ Create Project
Delete Projects
Displaying 6 items
Name
Description
Project ID
Domain Name
Enabled
Actions
Tdia_project
non
0f0e811f20a34d2ab5d4d262ac75ea22
Default
Yes
Manage Members
alt_demo
1cdfeef2edea4402b29d30342f5ed91b
demo
5a88c26709ab4f658a9db8c5b58ab75e
service
a6678e18082648388989ea0639b224eb
invisible_to_admin
c55454bae7824115ac62f4f1e404a3fc
admin
Bootstrap project for initializing the cloud.
d0d0d3c674764977839a89bfcbfe3065
Displaying 6 items

Then, we log in to the New- user account (Tdia_user) for network configuration before creating our instance:

Log in with user credentials.

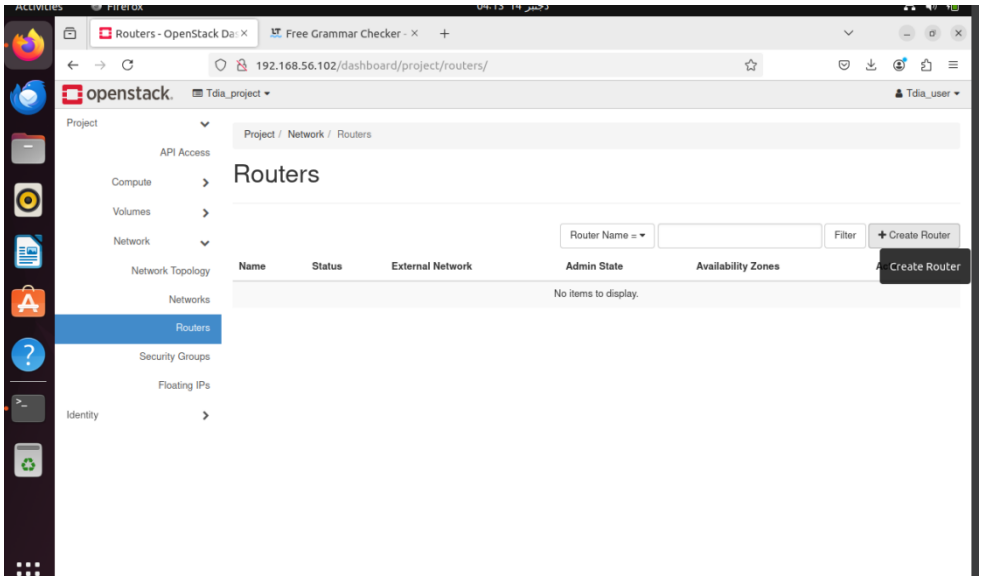




Next, we log in to the user account to configure the network before creating our instance.

3. Configure Network

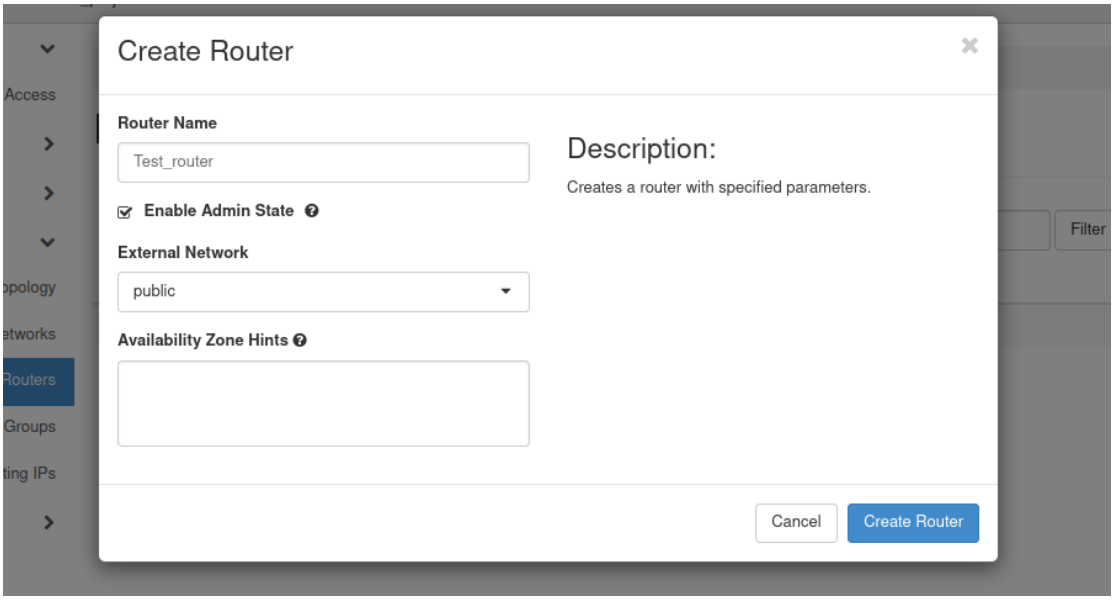
Step 1 : Creation of a router to interconnect our instance and public network beforehand

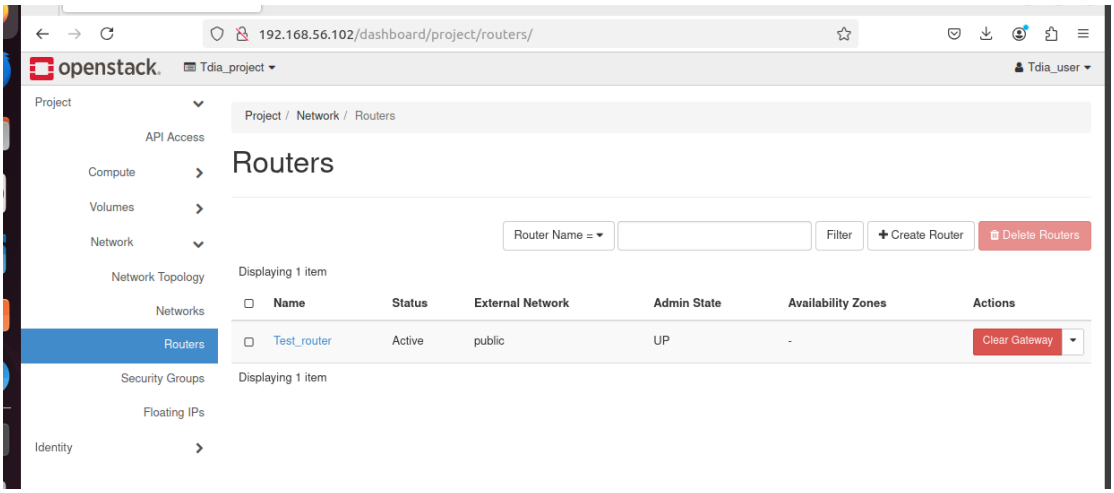


Creation of a new router:

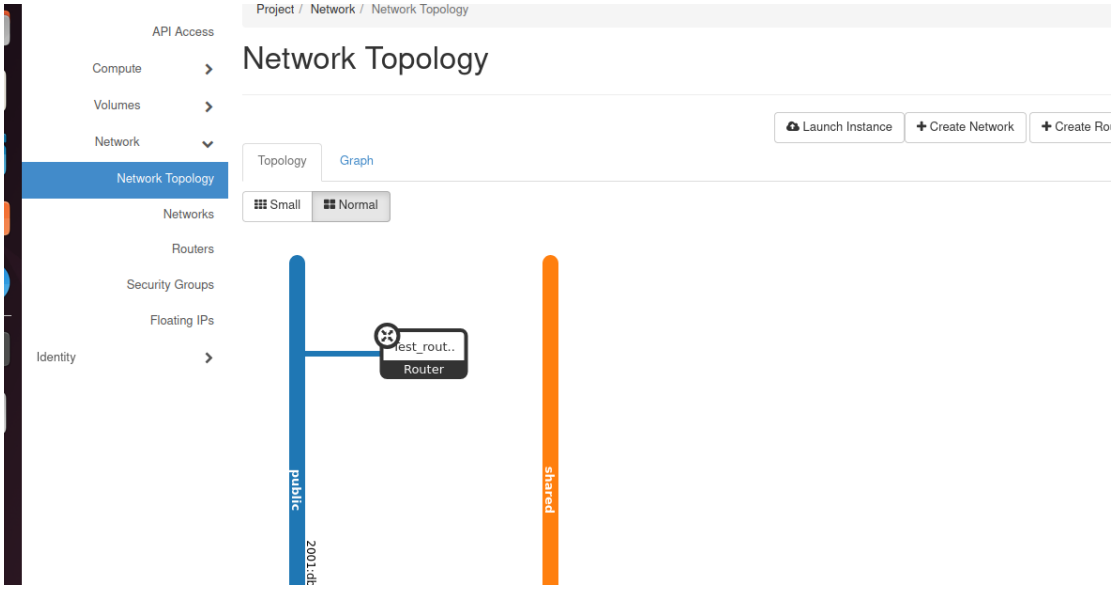
Name: test-router

- We will link it to the 'public' network.



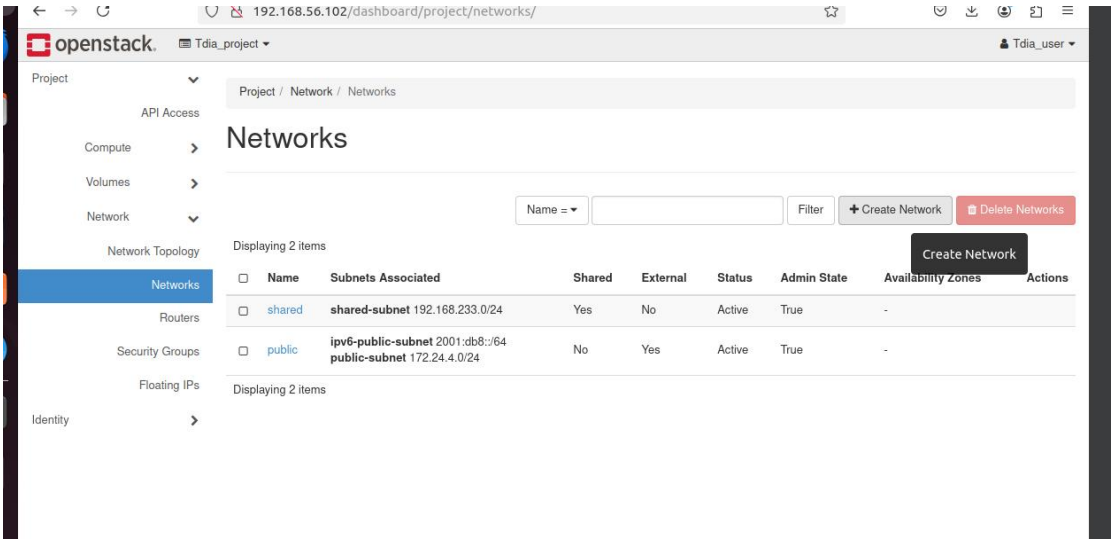


Our network creating a router :



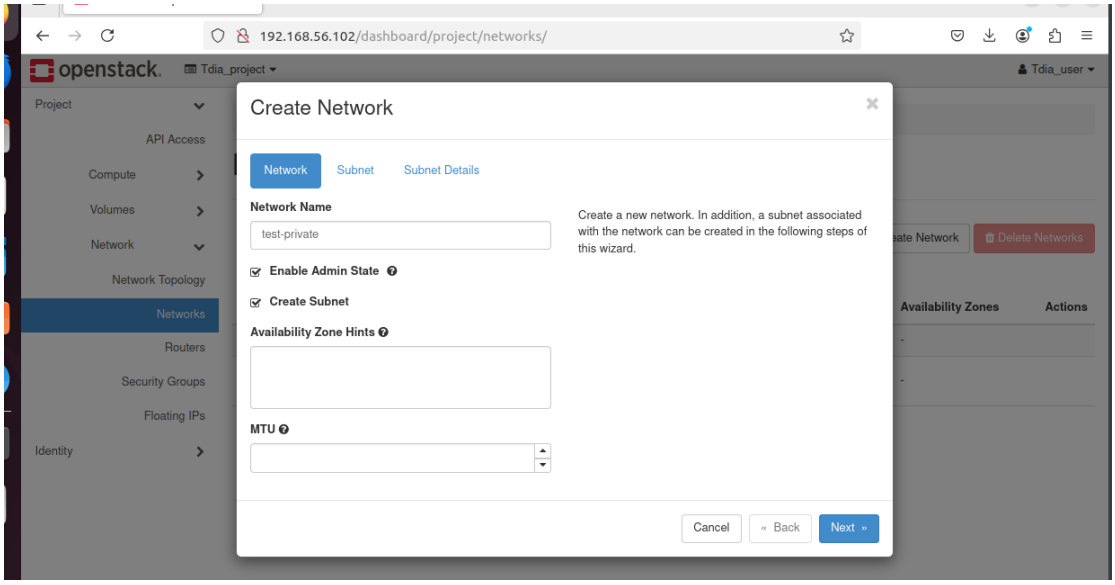
Step 2 : Creation of a private subnet

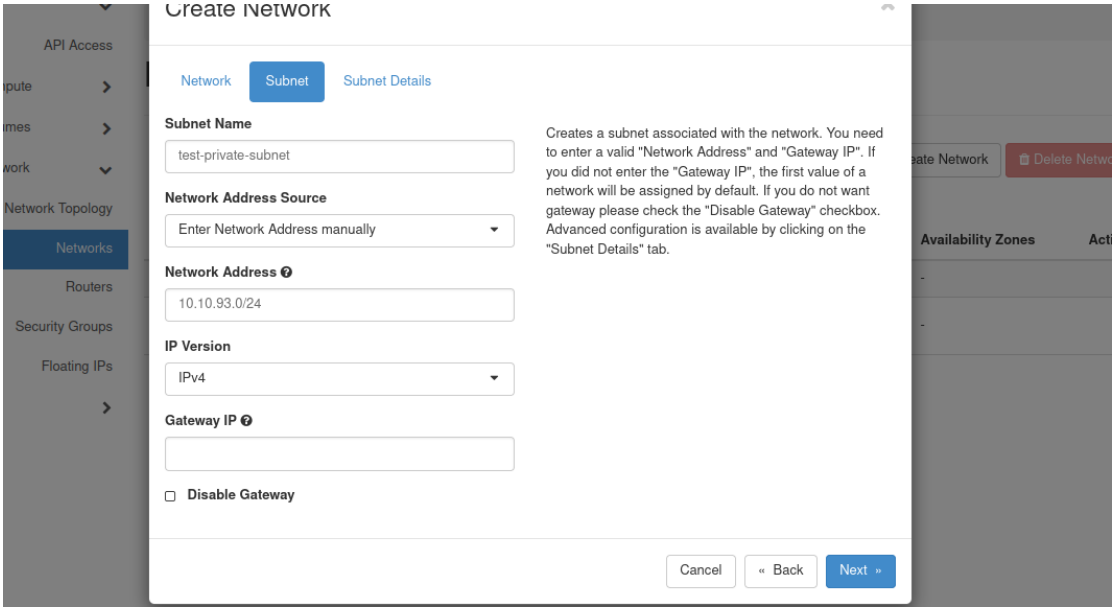
We assign it the name: test-private



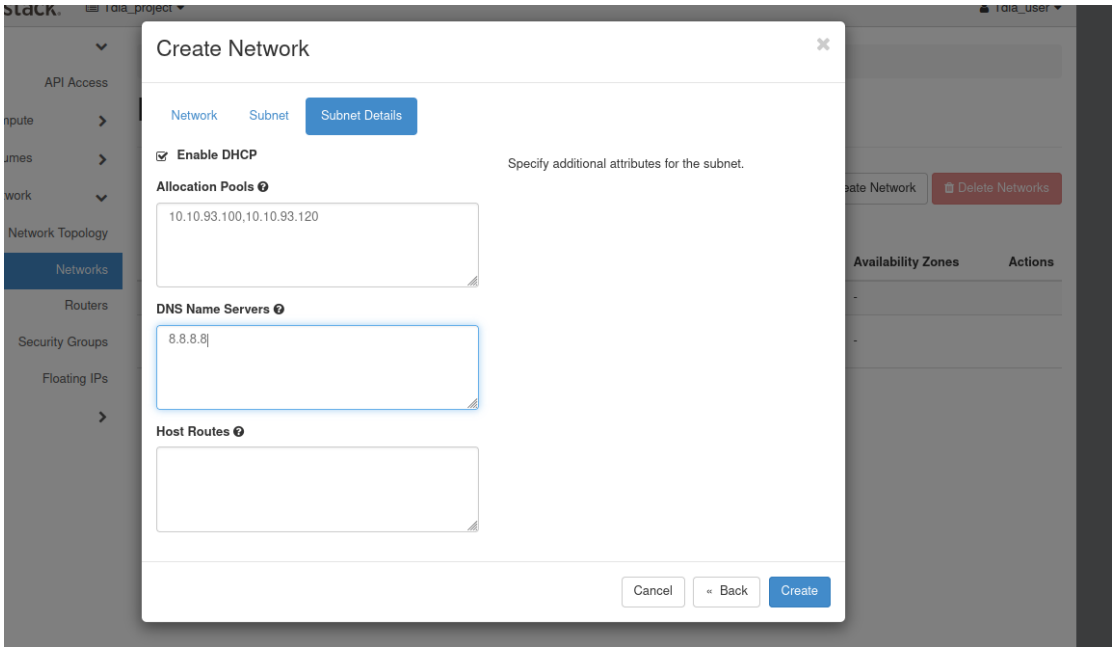
-Subnet name: test-private-subnet

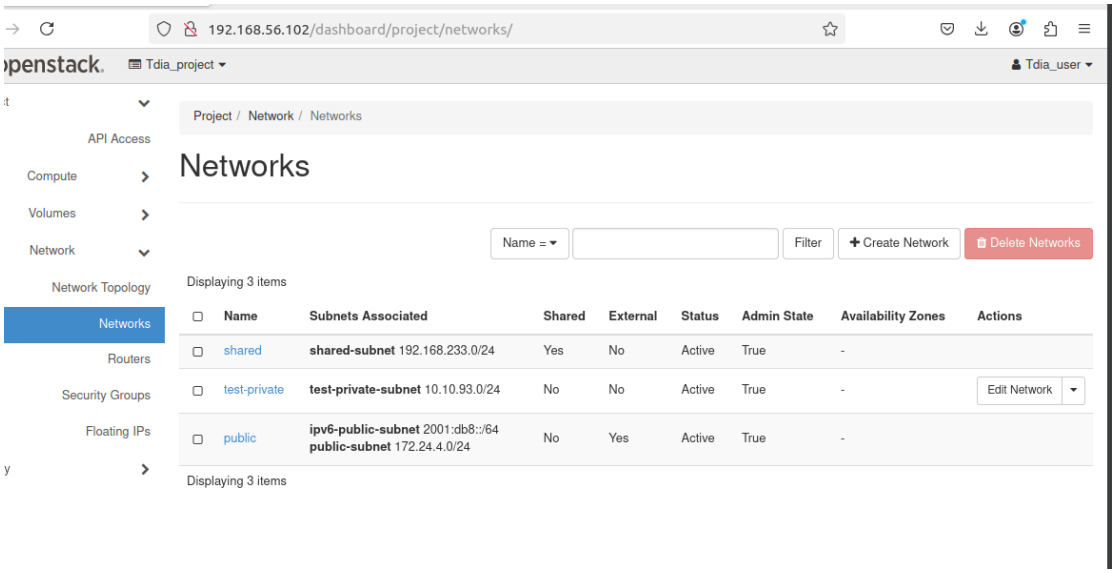
We specify the network address following the proposed syntax: **10.10.93.0/24** in IPv4, and for the gateway, we will leave it to be automatically selected.



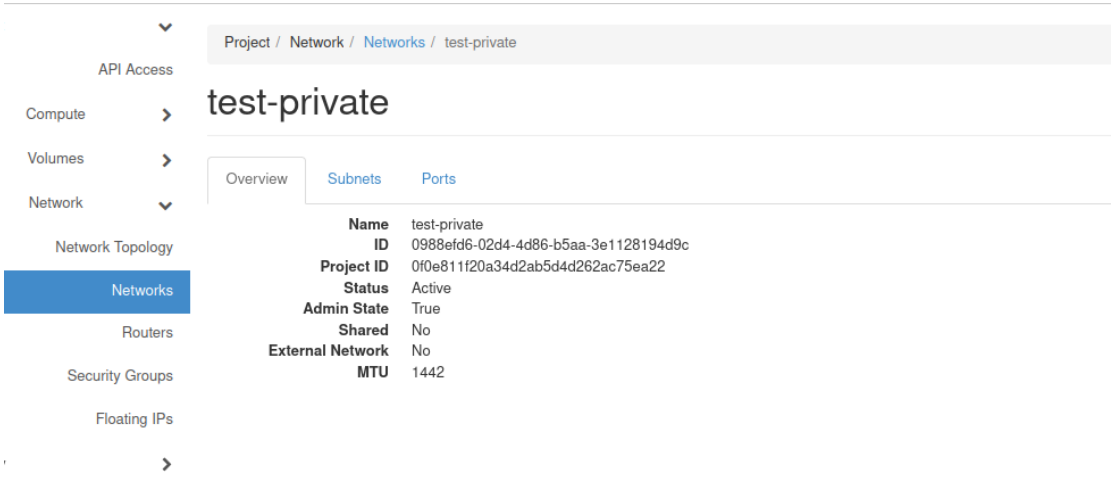


Then, we move on to the DHCP configuration:
For the address range, we provided: 10.10.93.100, 10.10.93.120.
For the DNS (Domain Name System), we selected Google's public DNS: 8.8.8.8.

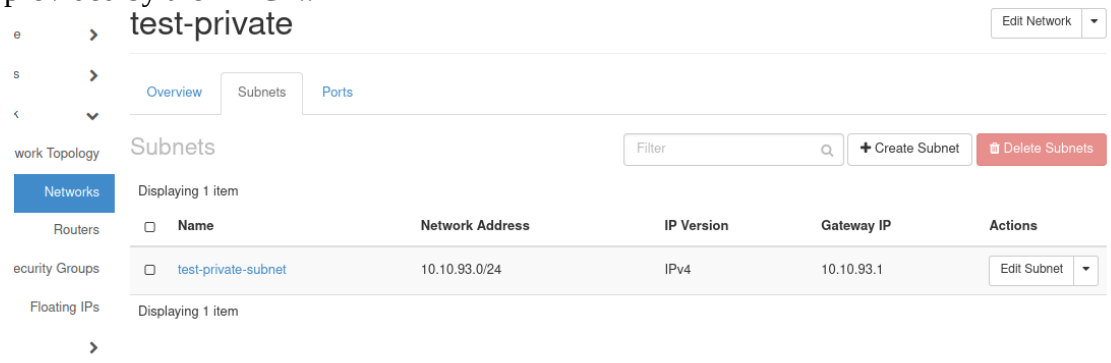




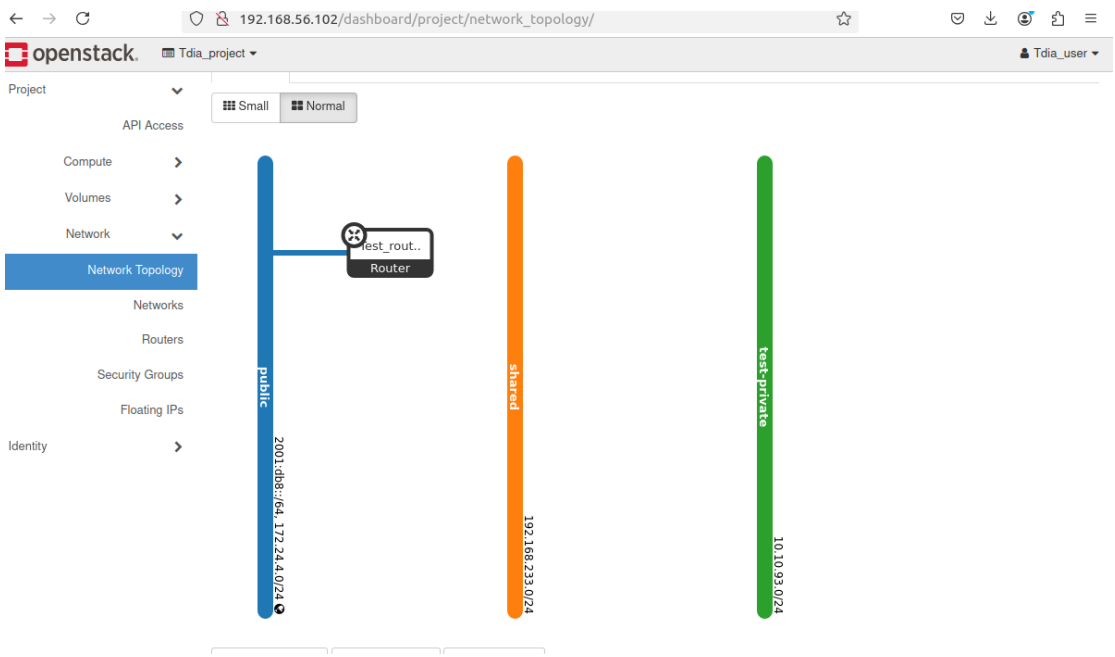
Overview of the network :



We can notice that even though we did not specify a Gateway IP, a Gateway IP is provided by the DHCP..

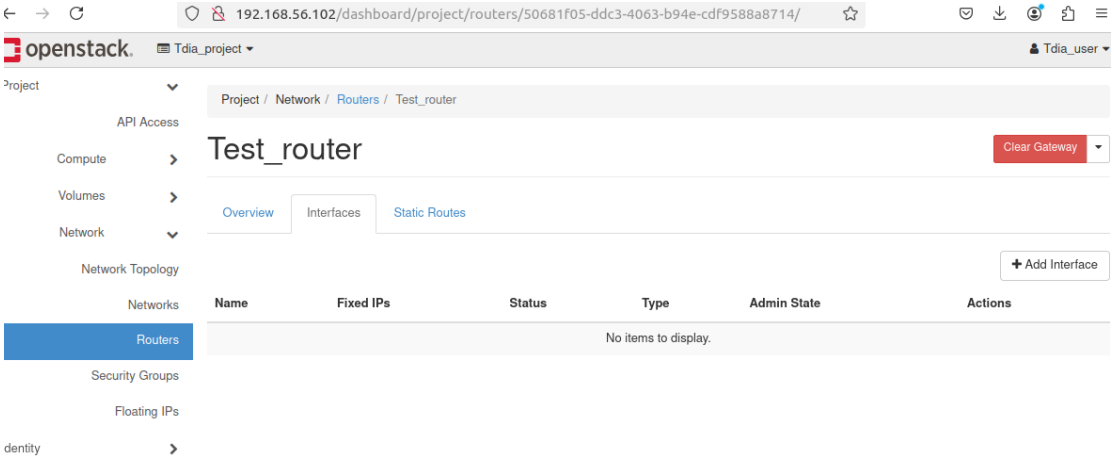


Our network after creating the subnet:

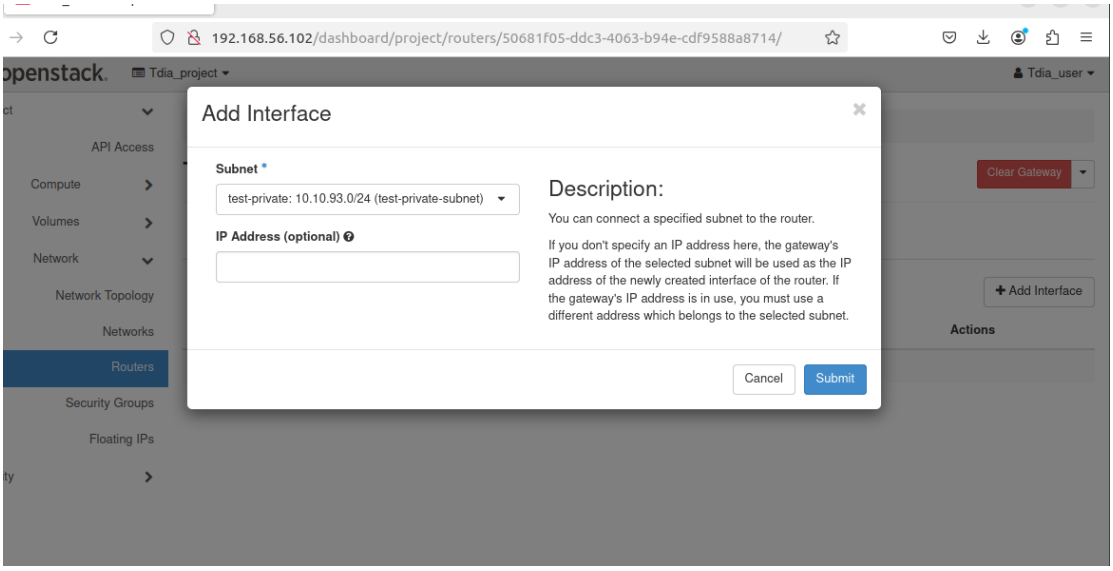


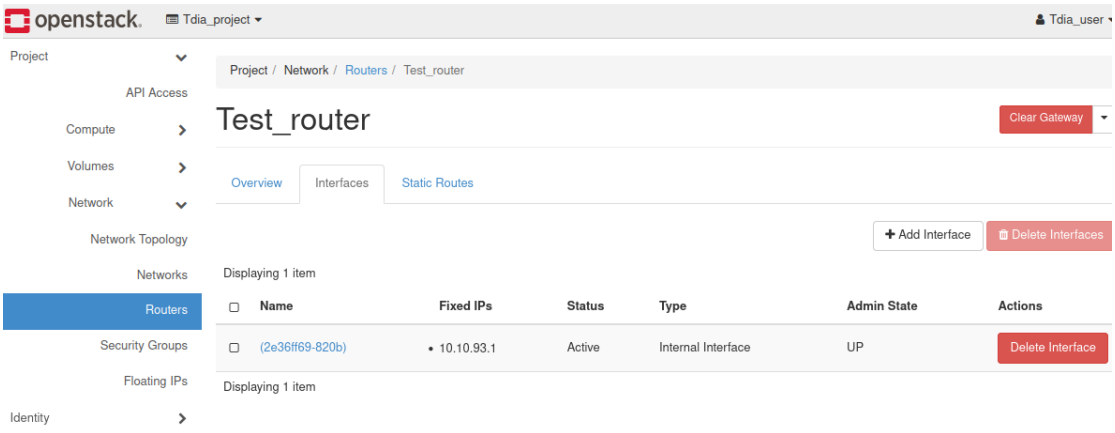
Step 3: Linking the private network to the router

First, we add an interface to our router.

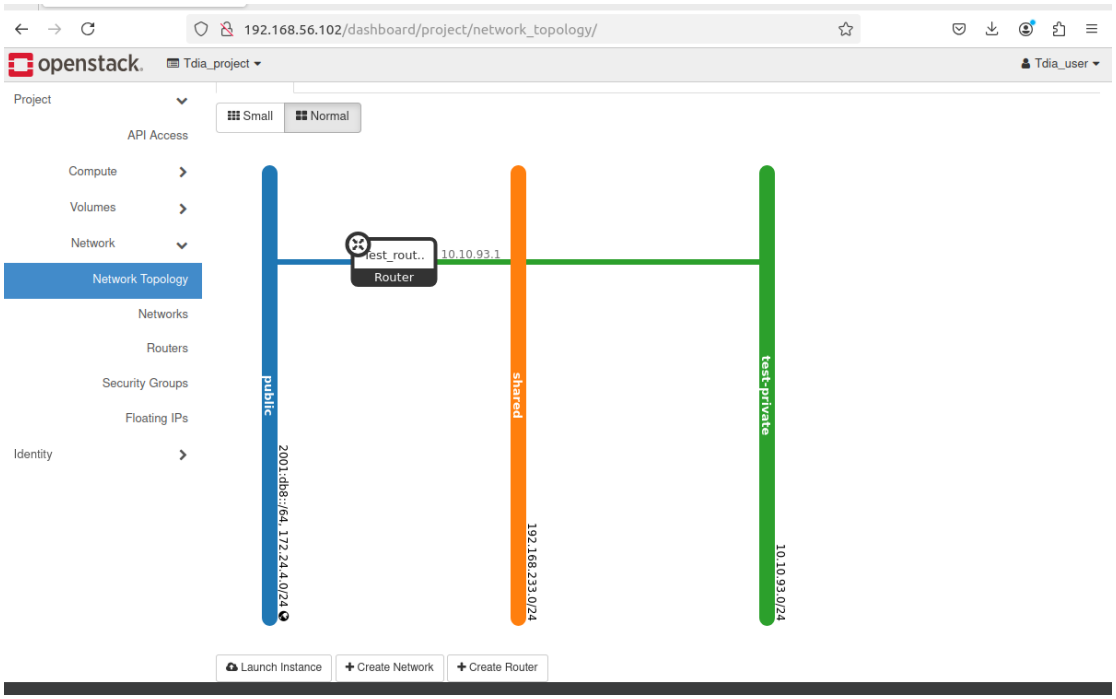


First, we select the subnet we just created:





Our network after the connection:
We can see that our private network is now linked to our public network via the router

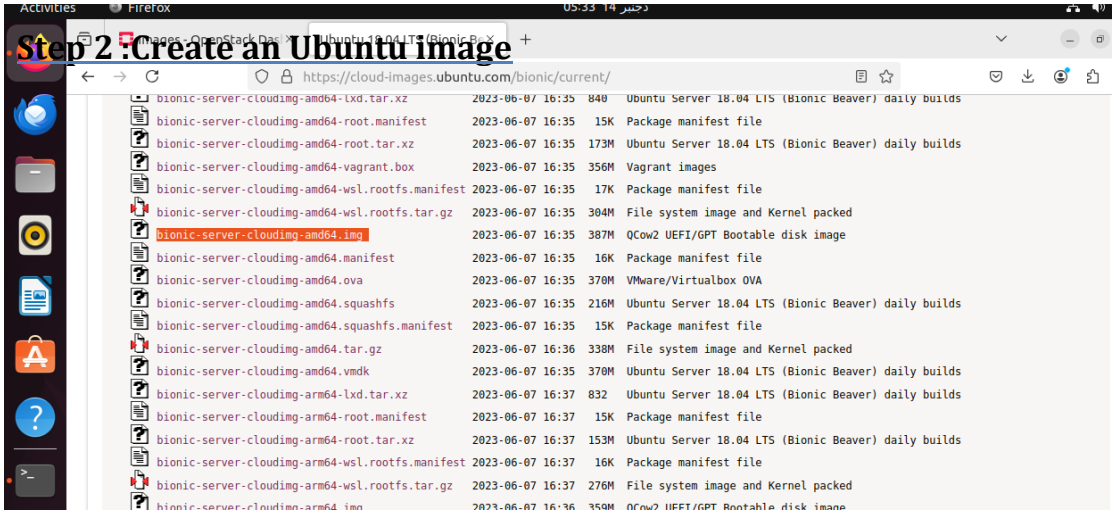


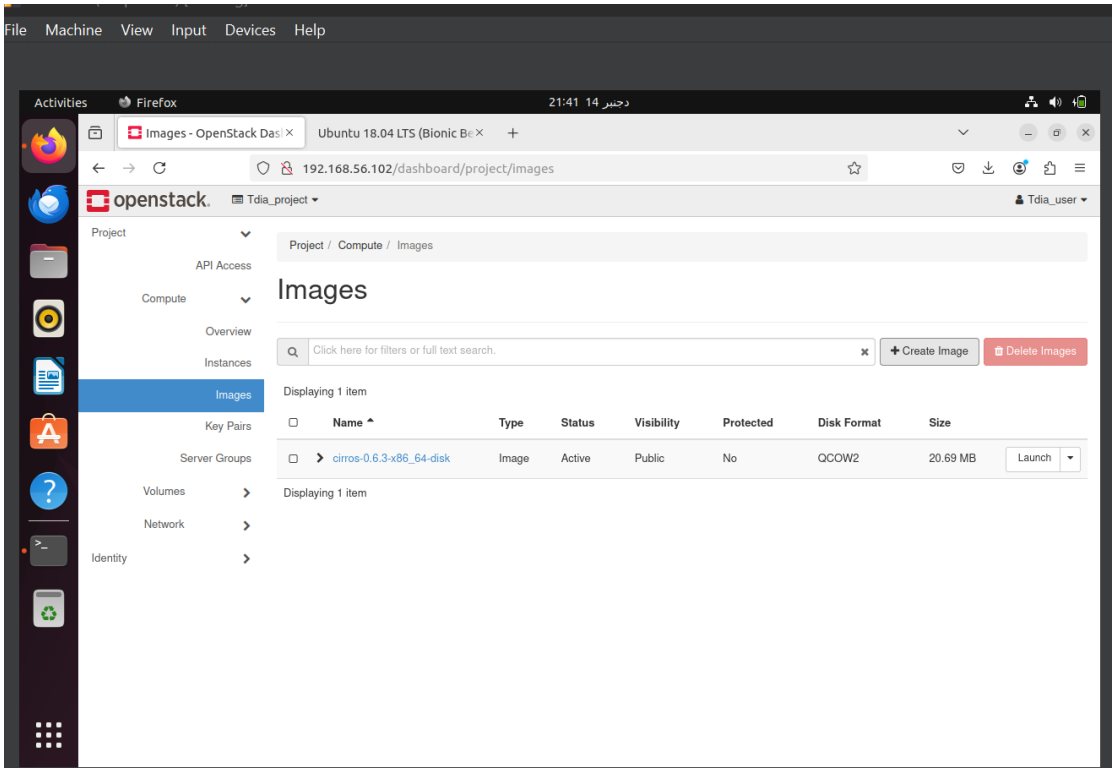
1. Create an Instance

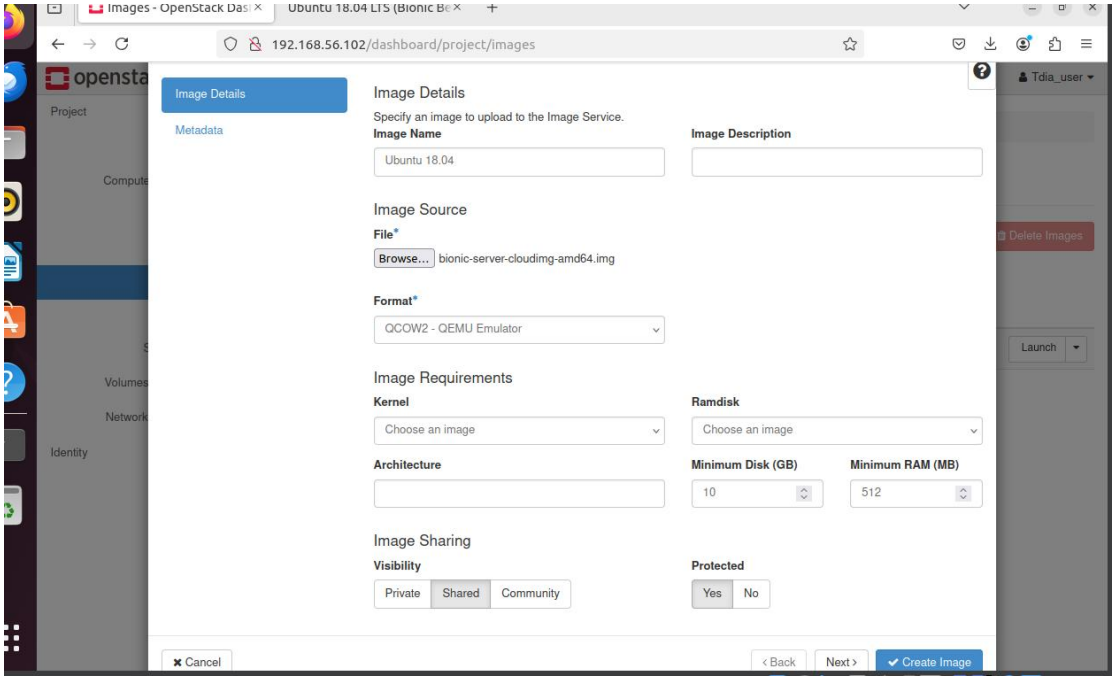
Step 1 : import Ubuntu image 18.04 made by Openstack:

[https://cloud-images.ubuntu.com/bionic/current/]

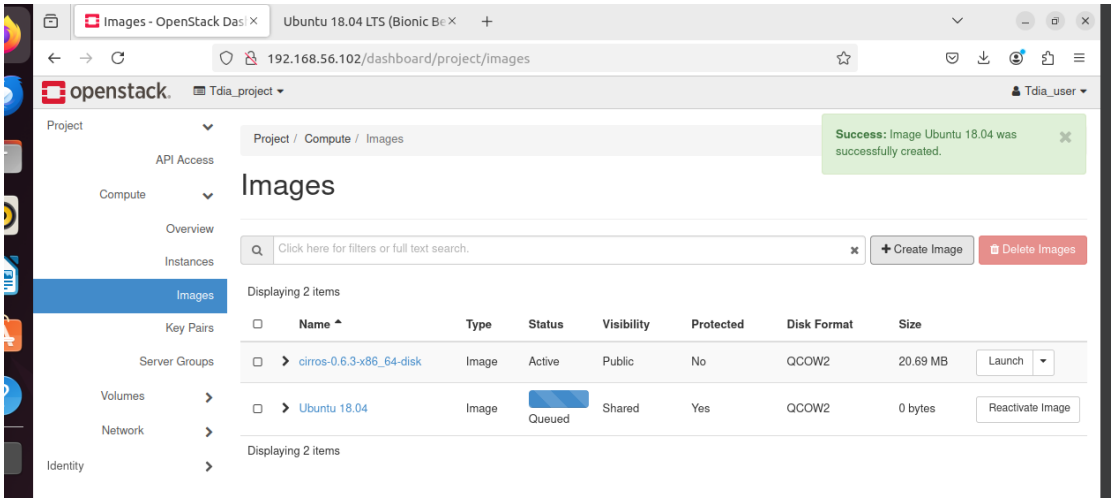
Step 2 :Create an Ubuntu image

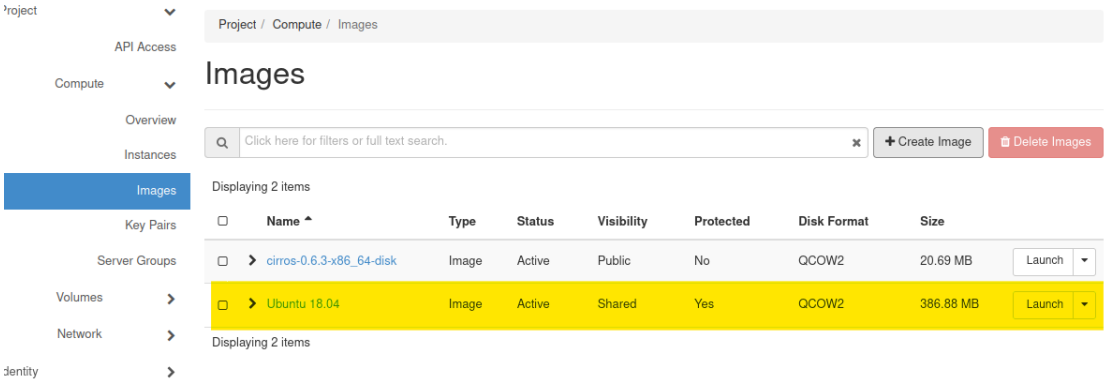






The Glance API is responsible for managing images in OpenStack.
The name of the image is set to: ubuntu 18.04.
The format of the image is specified as qcow2.
The minimum disk size is set to 10 GB.
The minimum RAM requirement is 512 MB.
The image is configured as shared between users and protected to prevent deletion.



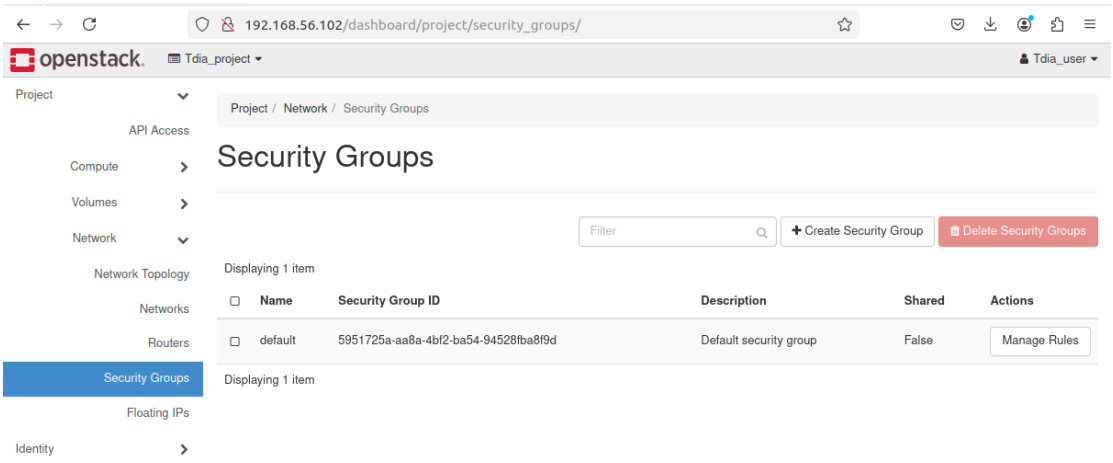


As you can see then image is created successffully.

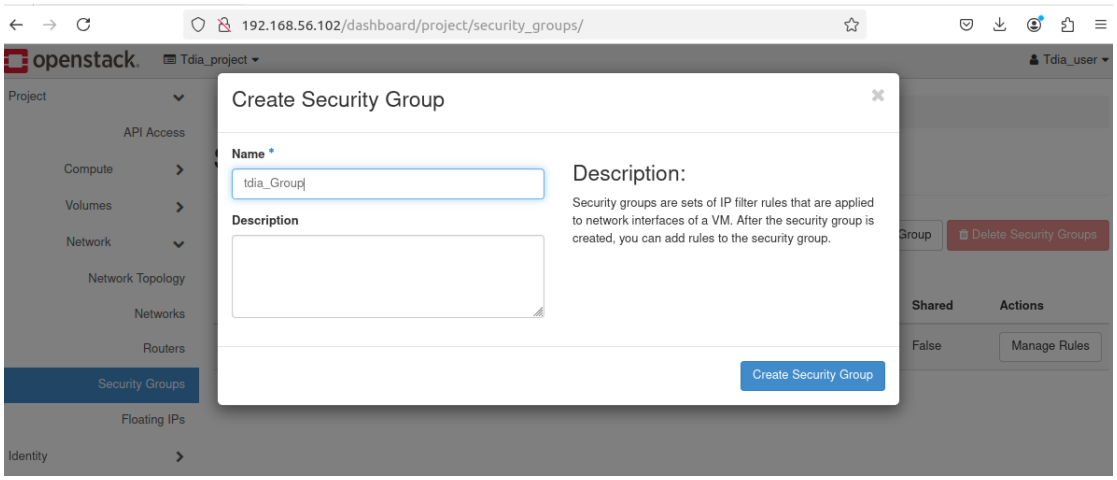
Step 3: Creating a Security Group

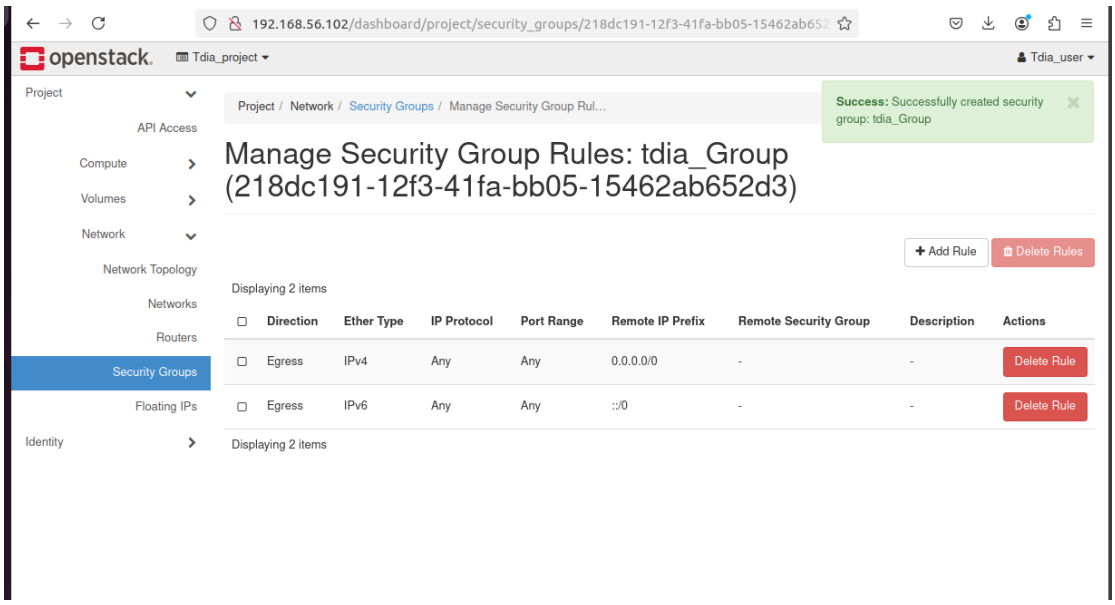
In OpenStack, a security group is a set of rules that control the incoming and outgoing network traffic for instances.

A) Navigate to “Project” , “Network” and then “Security Group”:



B) Define the Group Name: “tdia__Group” :





C) Add Rules to the Security Group:

by default in OpenStack, a security group allows all outbound traffic from a VM (Virtual Machine). This means that, unless specified otherwise, any traffic leaving the VM to the external network is permitted, regardless of the destination or protocol.

Outbound Traffic (Egress):
By default, all outgoing traffic from the VM is allowed. You don't need to define specific rules for outgoing traffic unless you want to restrict it.

Inbound Traffic (Ingress):
For incoming traffic, OpenStack security groups are more restrictive **by default**. Typically, only specific ports (such as SSH on port 22 or HTTP on port 80) are allowed, depending on the security group rules you define.

Specify inbound and outbound rules to allow or block specific types of traffic.

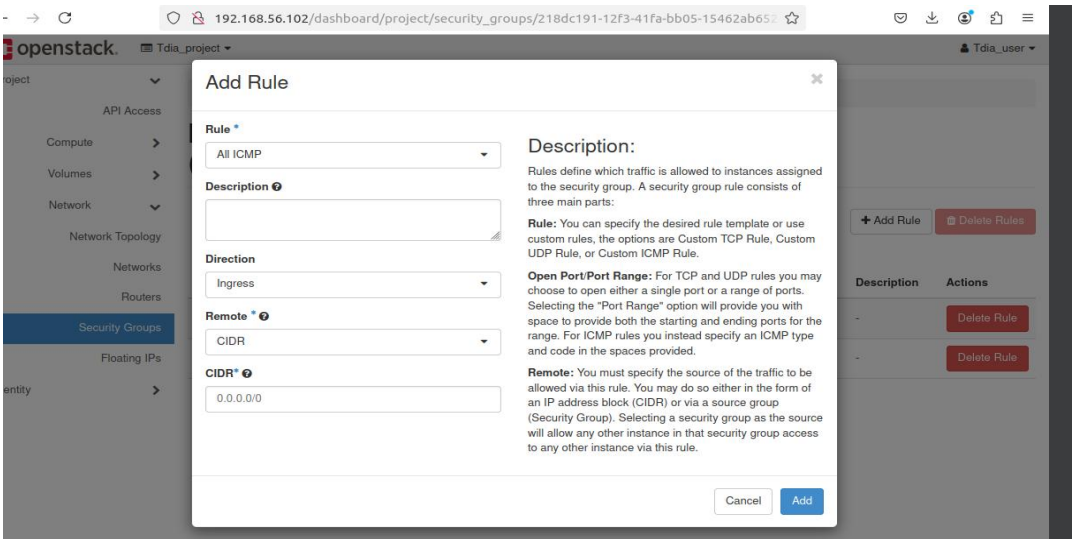
For example:

- Allow SSH (port 22) for remote access.
- Allow HTTP (port 80) and HTTPS (port 443) for web servers.
- Allow ICMP (ping) for diagnostics.

A) Add rule : ALL ICMP

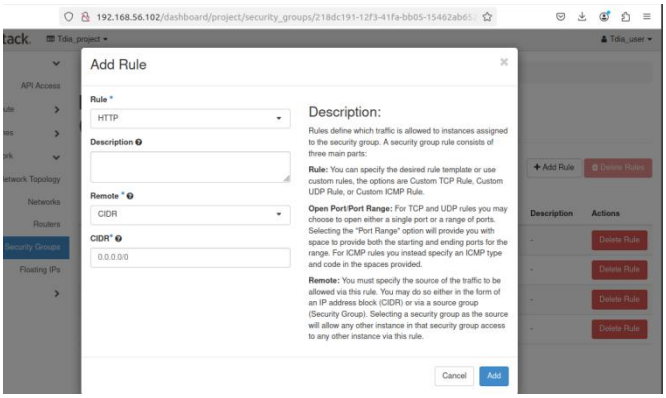
To add a rule that allows all ICMP traffic (which includes ping requests and replies), you need to configure the security group in OpenStack. ICMP traffic is used for diagnostic purposes, like using the ping command to check network connectivity.

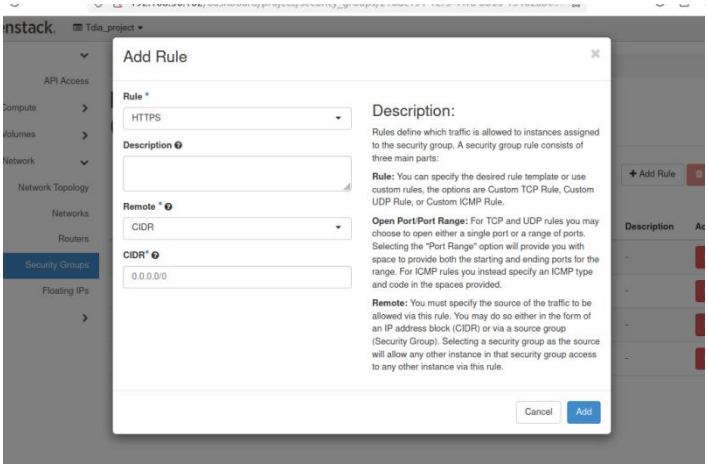
This can be useful for network diagnostics but should be used carefully in production environments as it can expose your VMs to unnecessary network traffic.



B) Add rule : HTTP and HTTPS

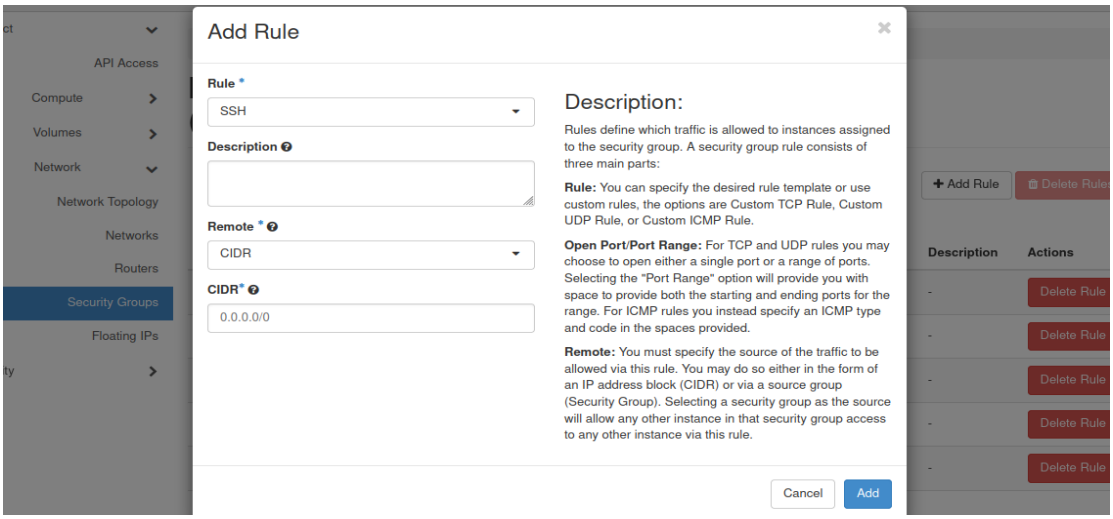
These rules are commonly used to allow web servers to communicate with users over the internet.



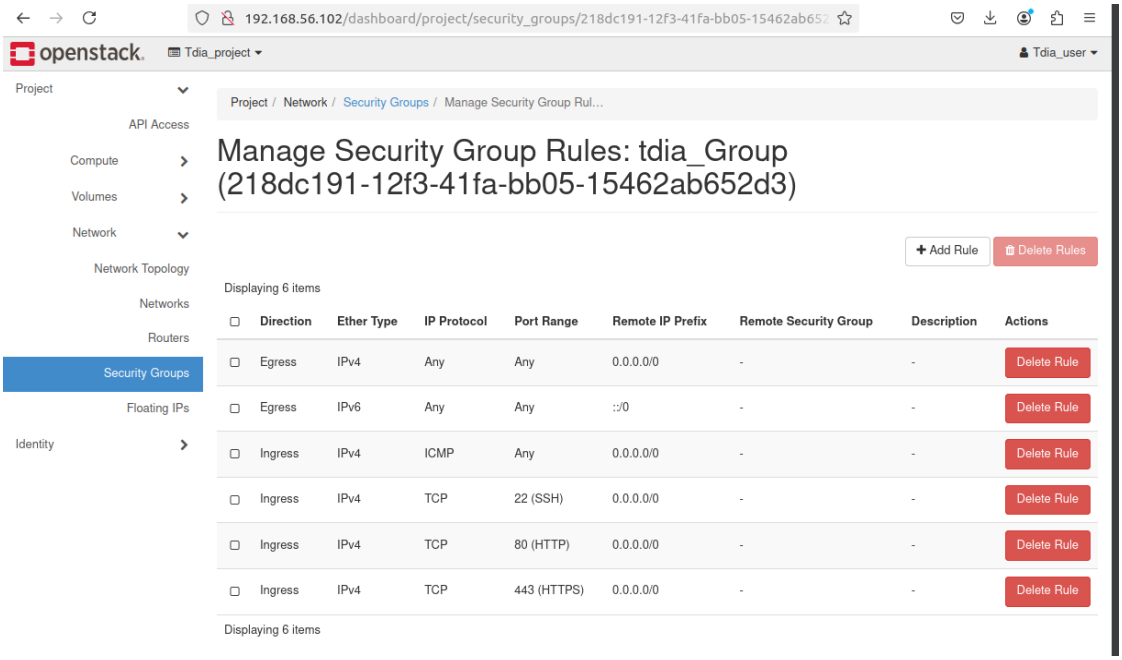


C) Add rule : SSH

which is very important for us to be able to administer our machine later on.



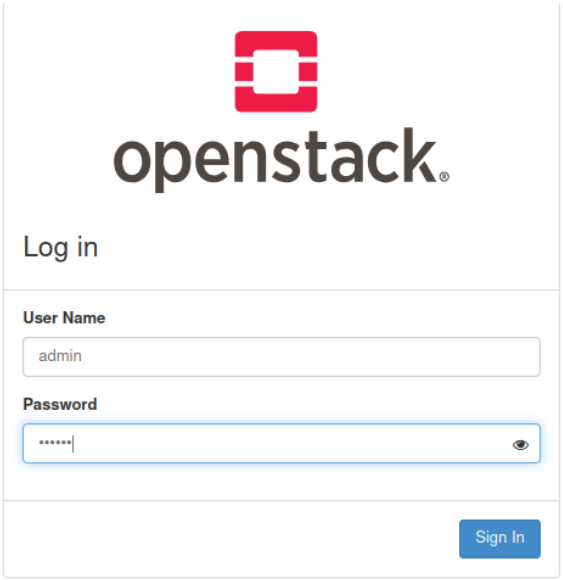
The set of rules created



Step 7: Creating a New Flavor in OpenStack

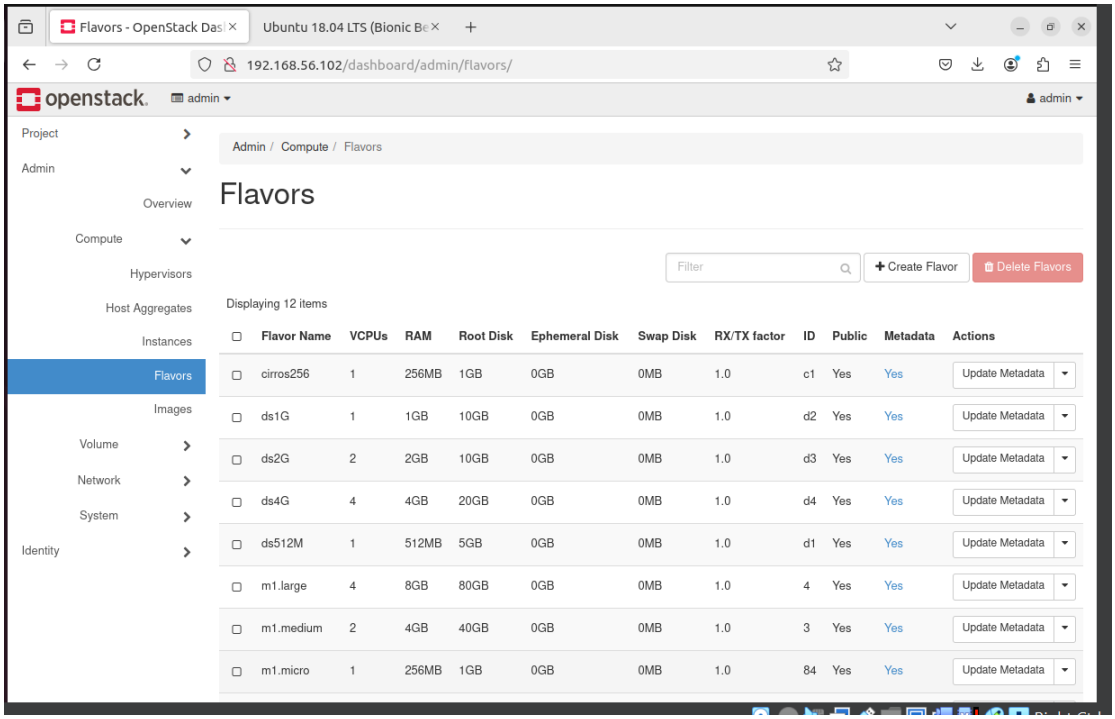
Creating a flavor in OpenStack allows you to define the specifications (vCPUs, RAM, disk, etc.) for virtual machines (VMs) that users can launch.

Then we connect again as admin:



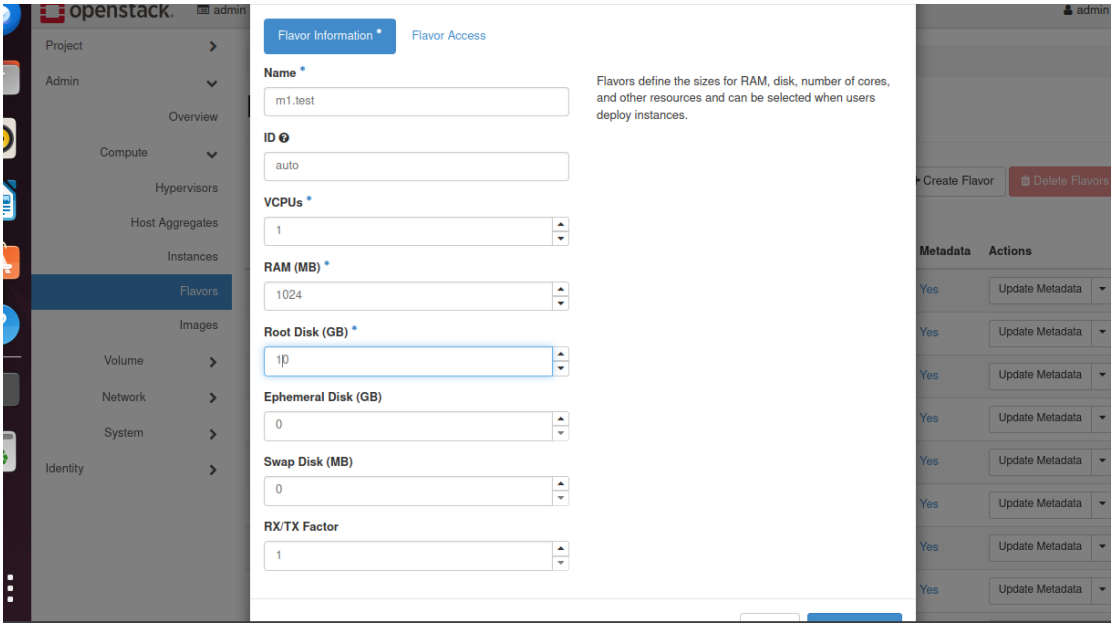
Creating a new flavor by specifying the resources we want:
Name: m1.test
VCPUs: 1
RAM: 1024 MB
ROOT DISK: 10 GB
(while respecting the characteristics we specified during the creation of our image)

Navigate to falvor Section :
Click on the "Admin" tab (or "System", depending on your OpenStack version).
From the side menu, select "Flavors" under the "Compute" category.



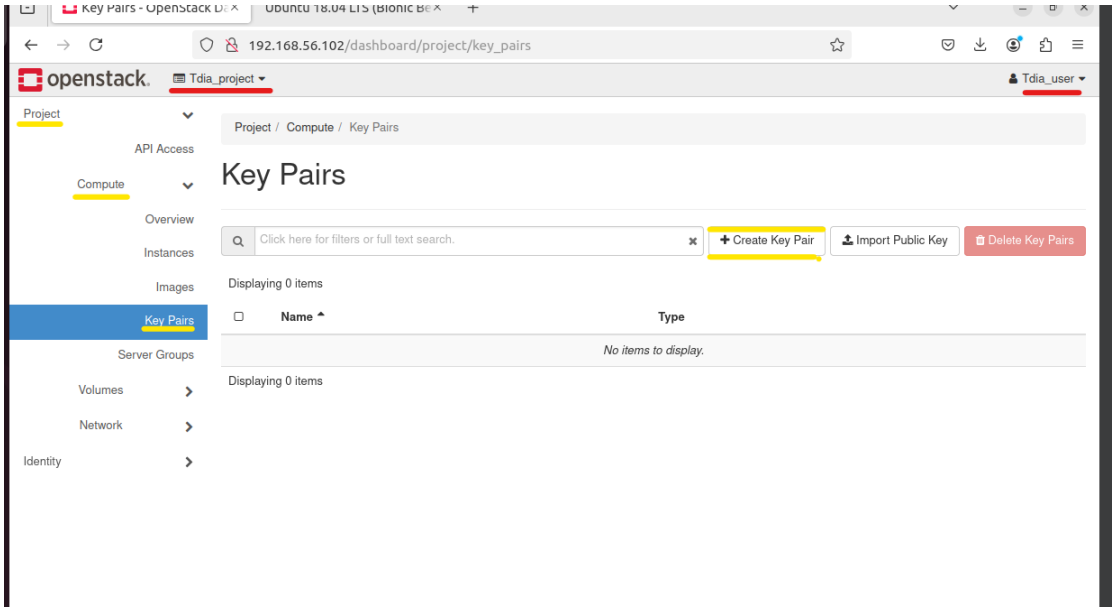
Fill in the New Flavor Details:

- **Name:** Provide a unique name for the flavor (e.g., `m1.test`).
- **ID:** You can assign a specific ID or leave it blank for the system to auto-generate one.
- **vCPUs:** Enter the number of virtual CPUs for this flavor (e.g., 1).
- **RAM (MB):** Specify the amount of RAM in megabytes (e.g., 1024 for 1 GB).
- **Root Disk (GB):** Enter the size of the root disk in gigabytes (e.g., `10`).
- **Ephemeral Disk (GB):** Specify the size of the ephemeral disk, if any (optional).
- **Swap Disk (MB):** Enter the swap disk size in megabytes (optional).
- **RX/TX Factor:** This is used for bandwidth scaling (leave as `1.0` if unsure).



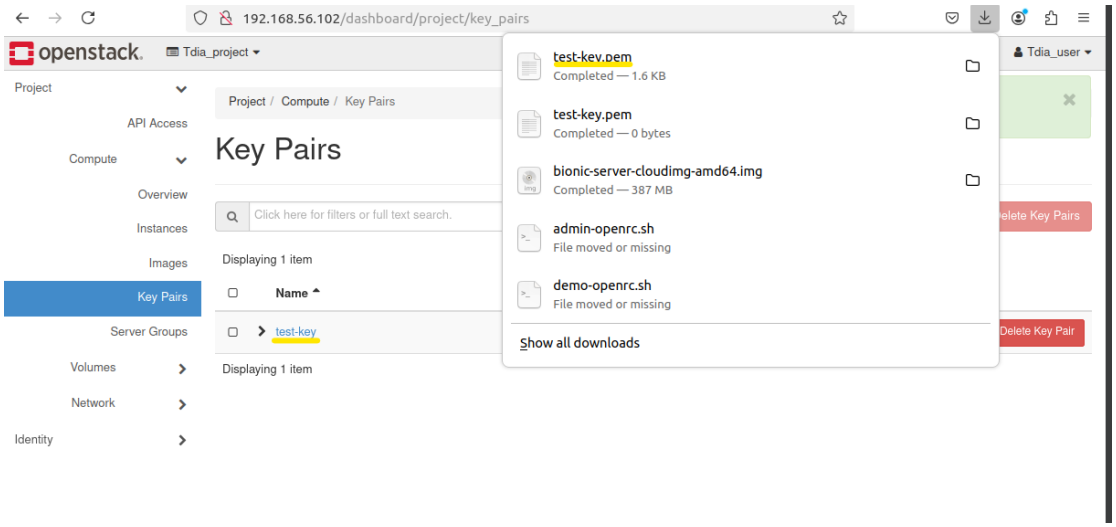
Etape 8 : création d'une paire de clé

Connect as Tdia_user and navigate to “**key Pairs**” section

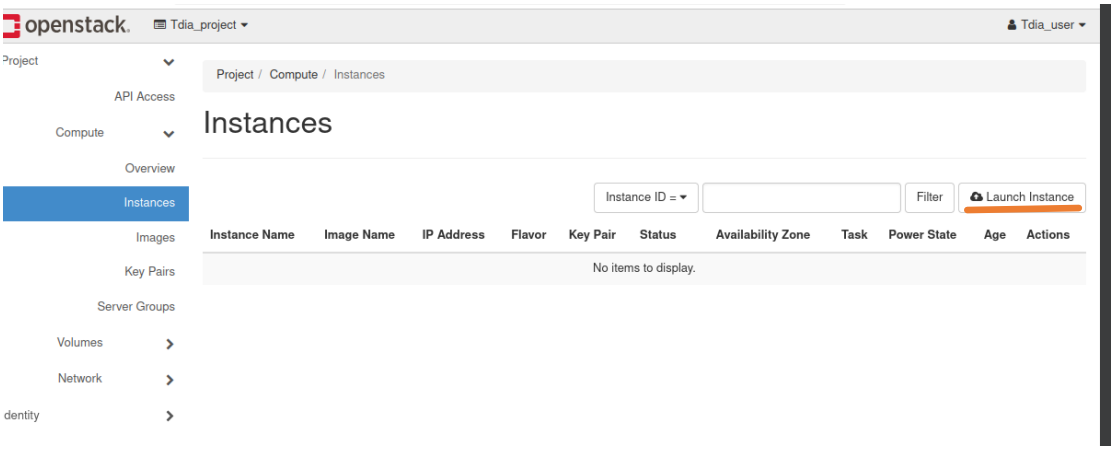


The SSH key test-key that we are asked to create, then we retrieve the generated key to use during SSH connection

As we can see, the key is generated automatically and downloaded to your machine.

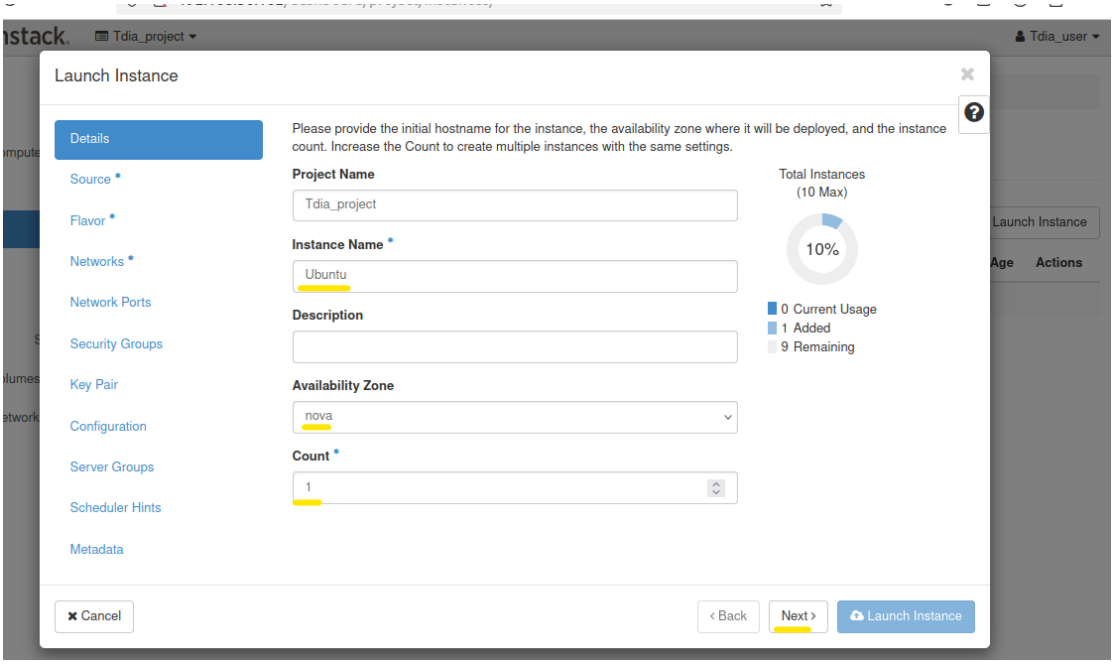


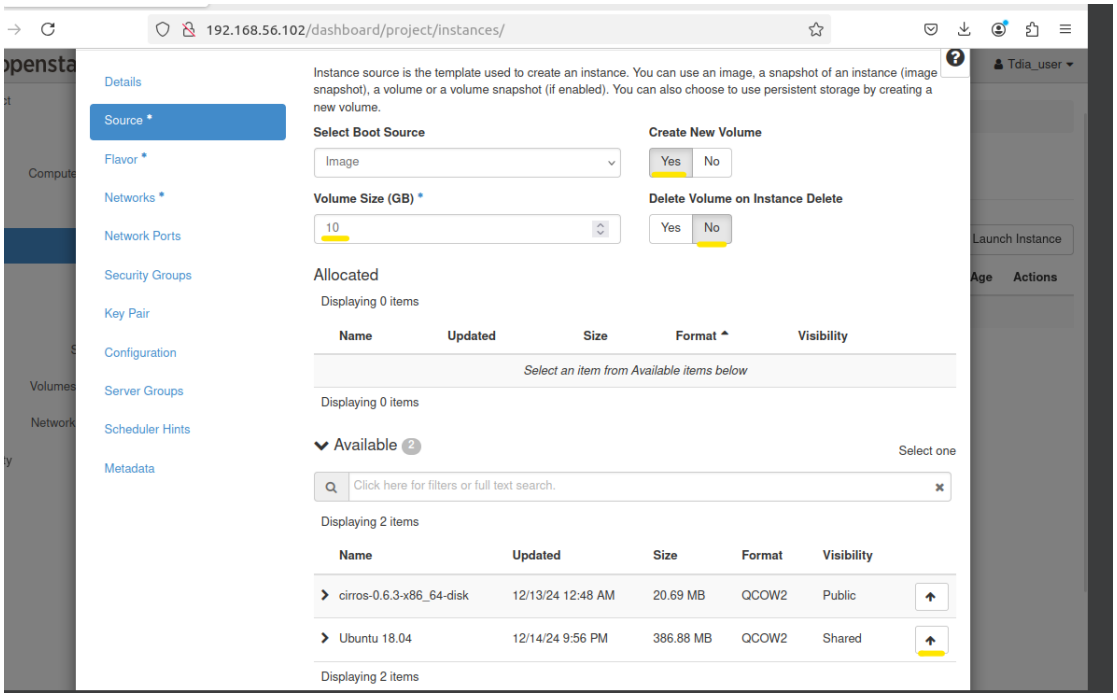
Step 9: Creating the instance



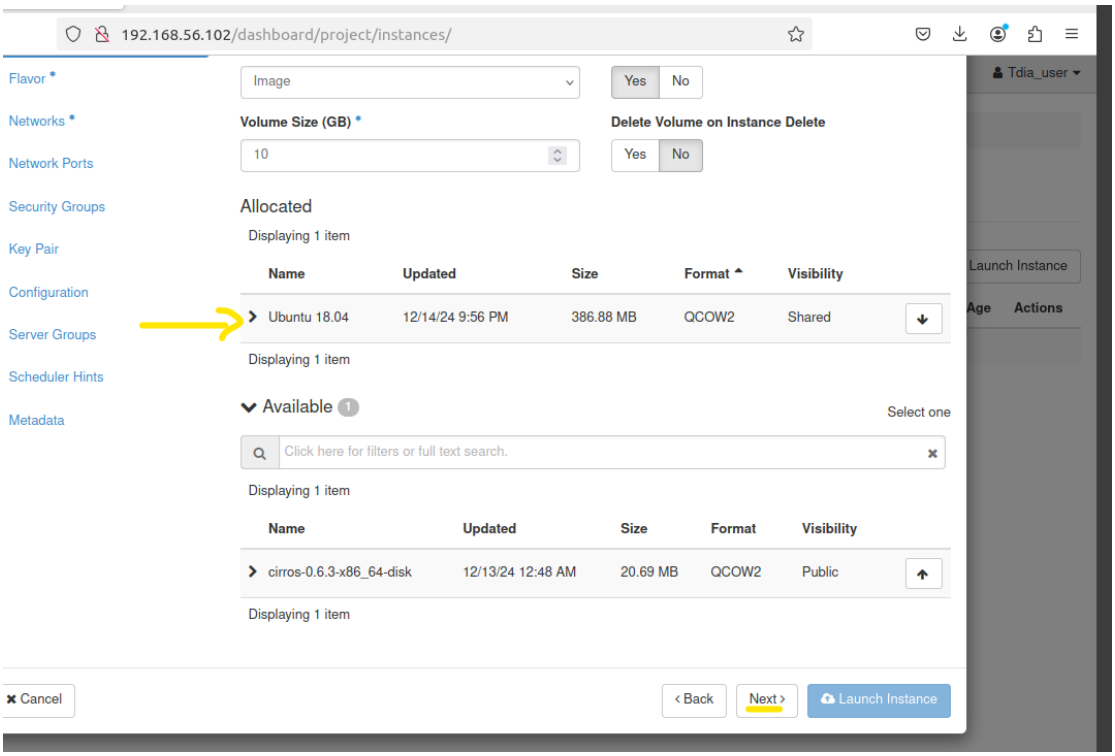
Set Instance Details [Project Name : The scope of your Instance,
Name : Ubuntu ,

Description : (optional)]

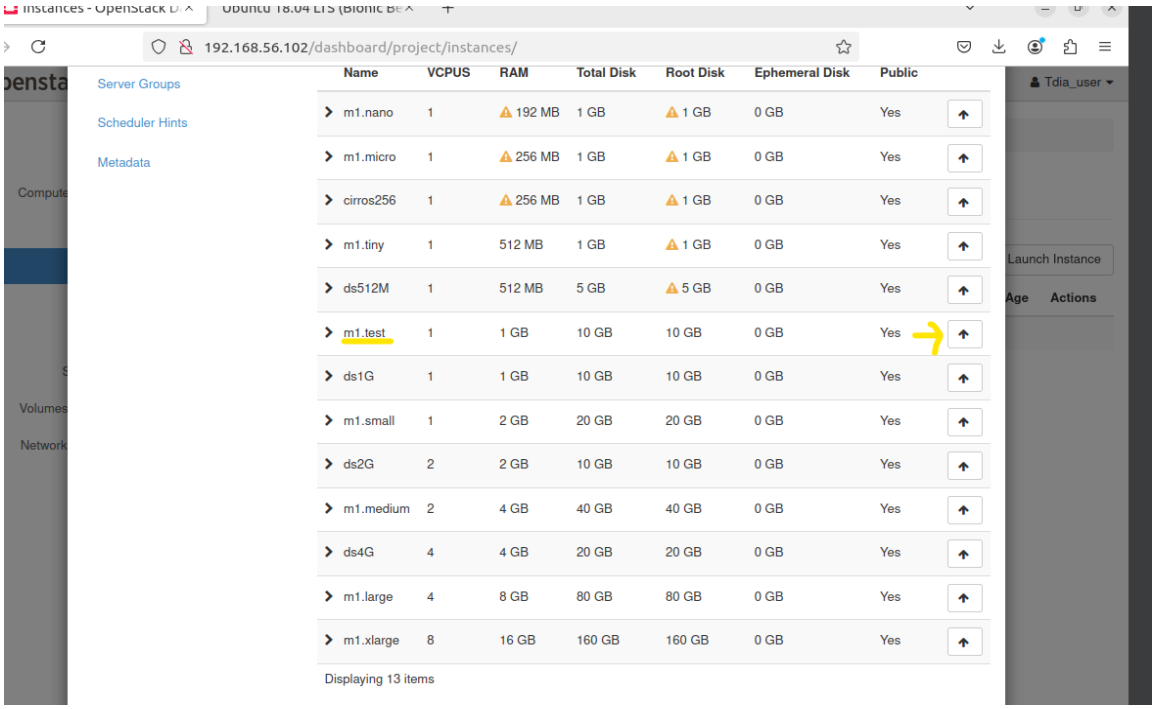




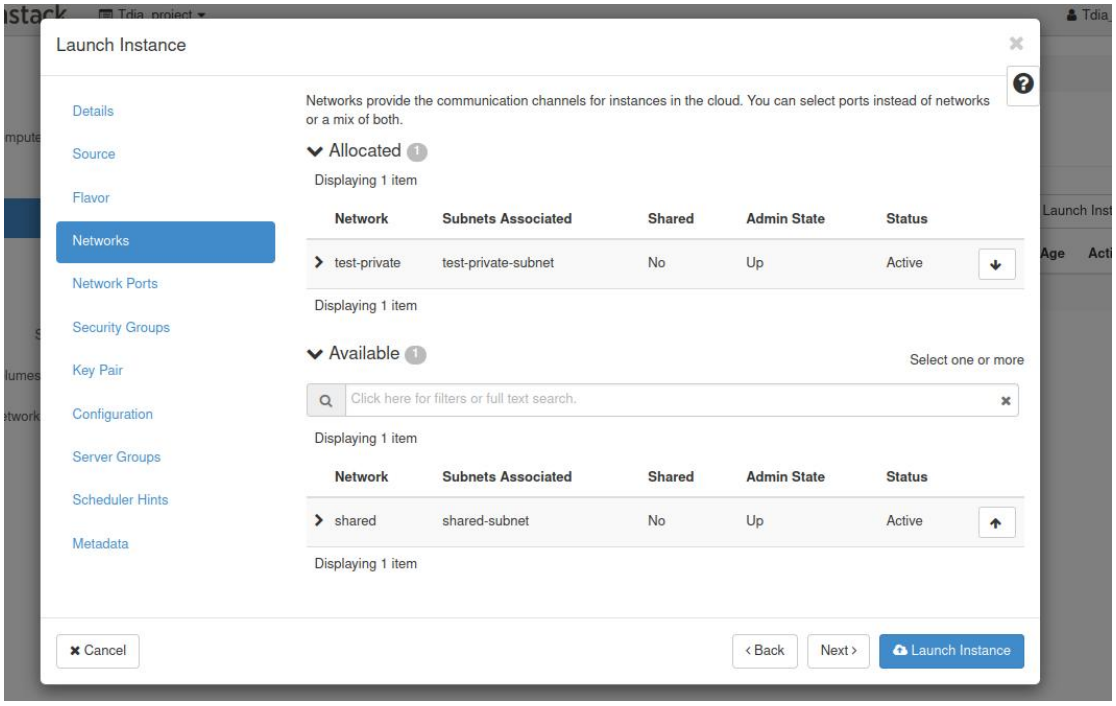
Click on the up-arrow associated with the Ubuntu image we downloaded earlier.



Then select the flavor we created for this image: m1.test

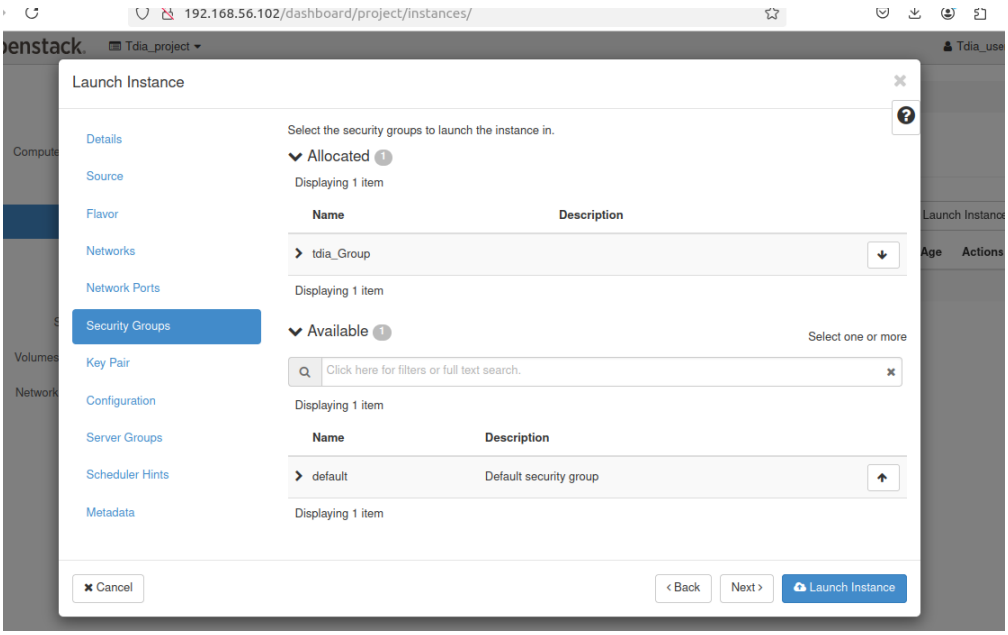


Then we select the private network we created: test-private

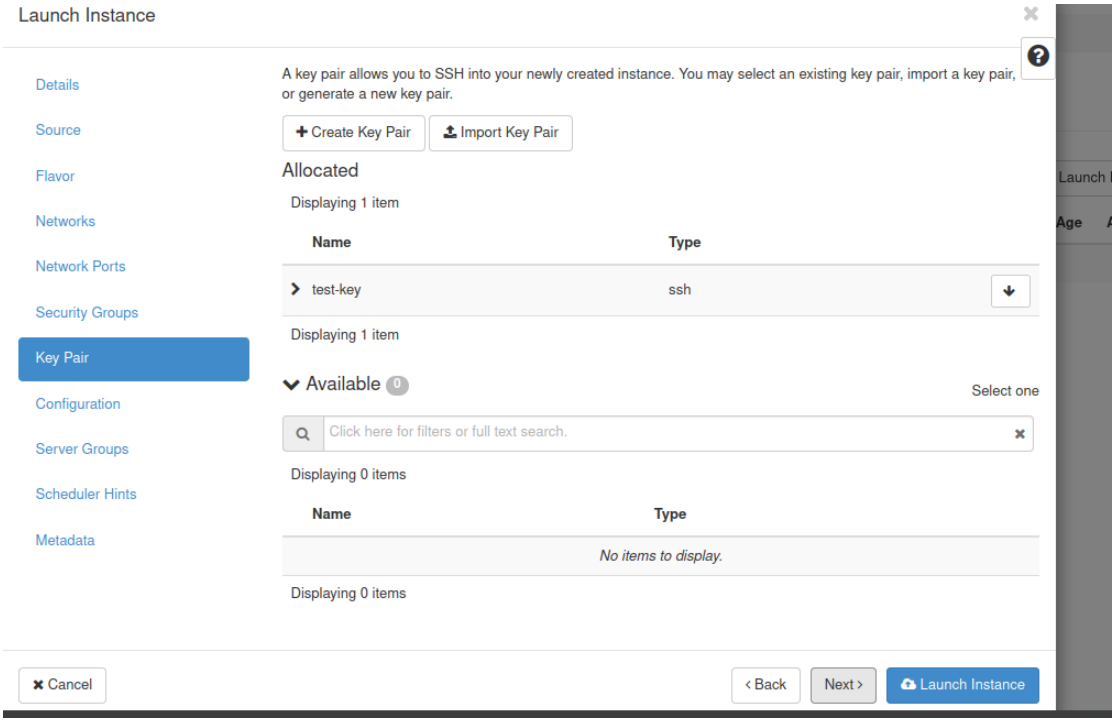


(#!Skip **Network Post** part)

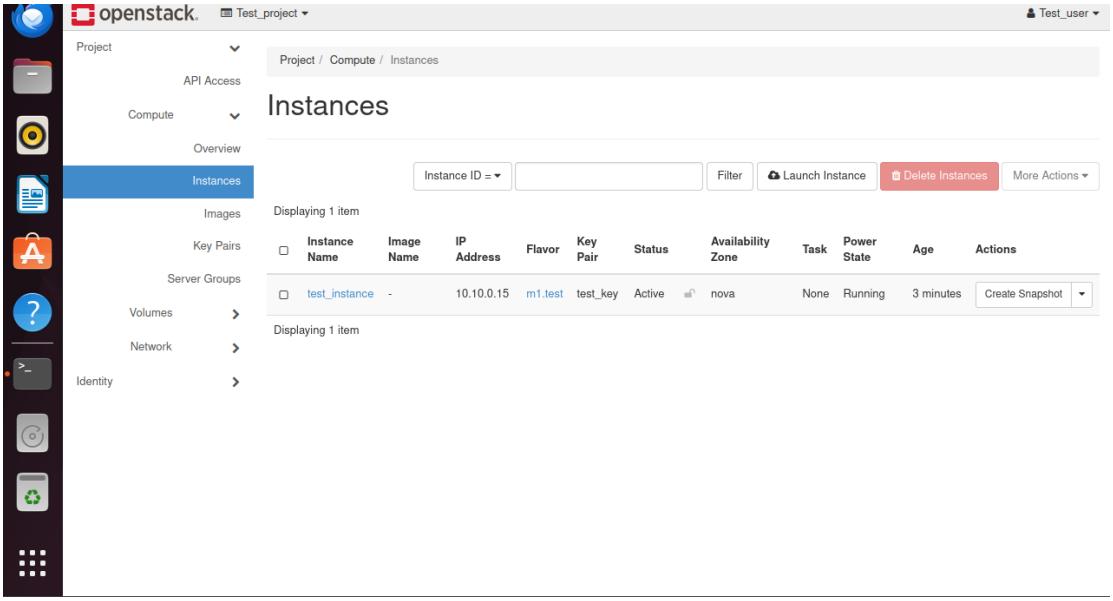
Add the group we created before :

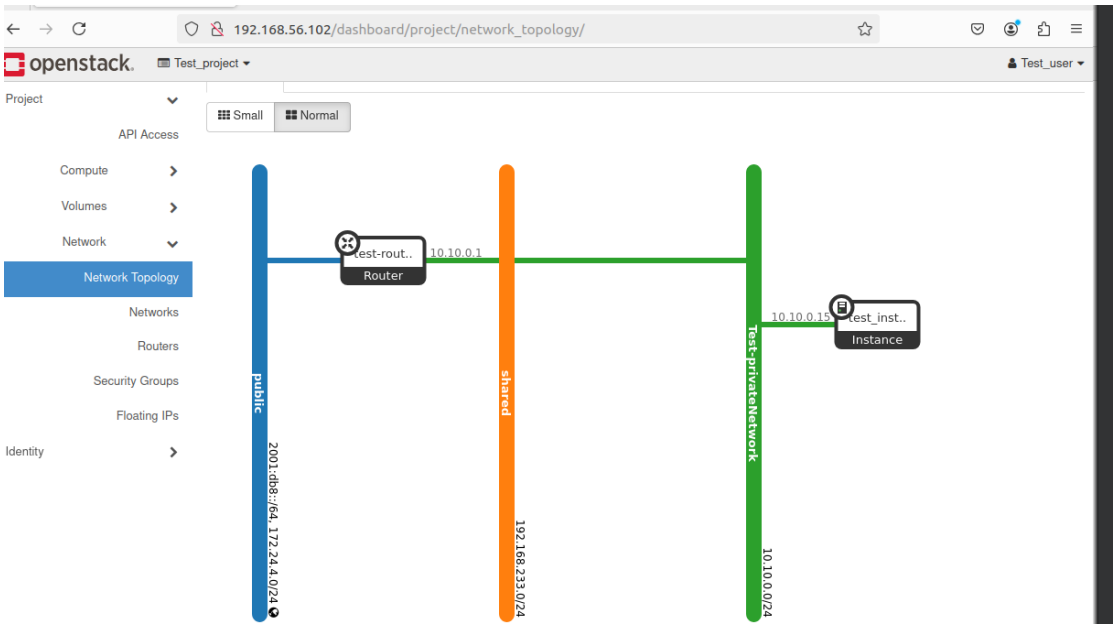


Select the Key ‘key-test’ :



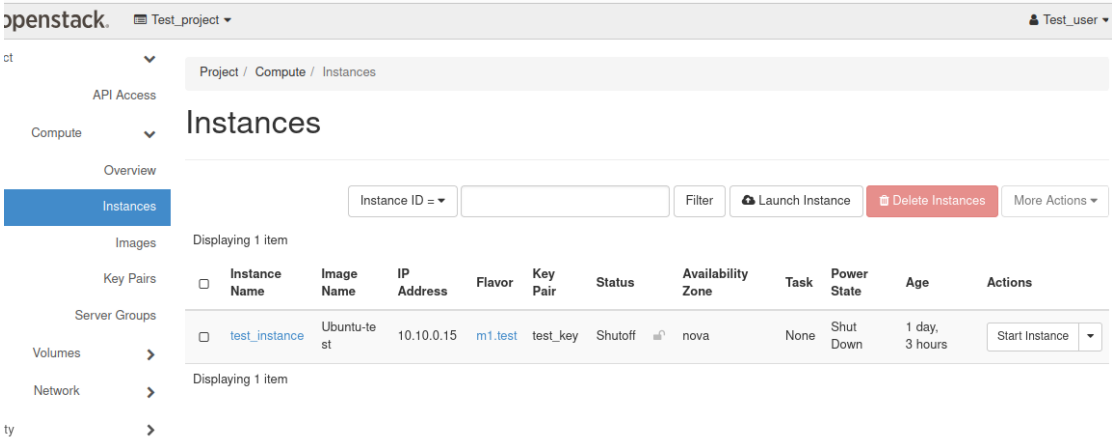
And then skip all other steps (keep them in default values) and lunch the instance.



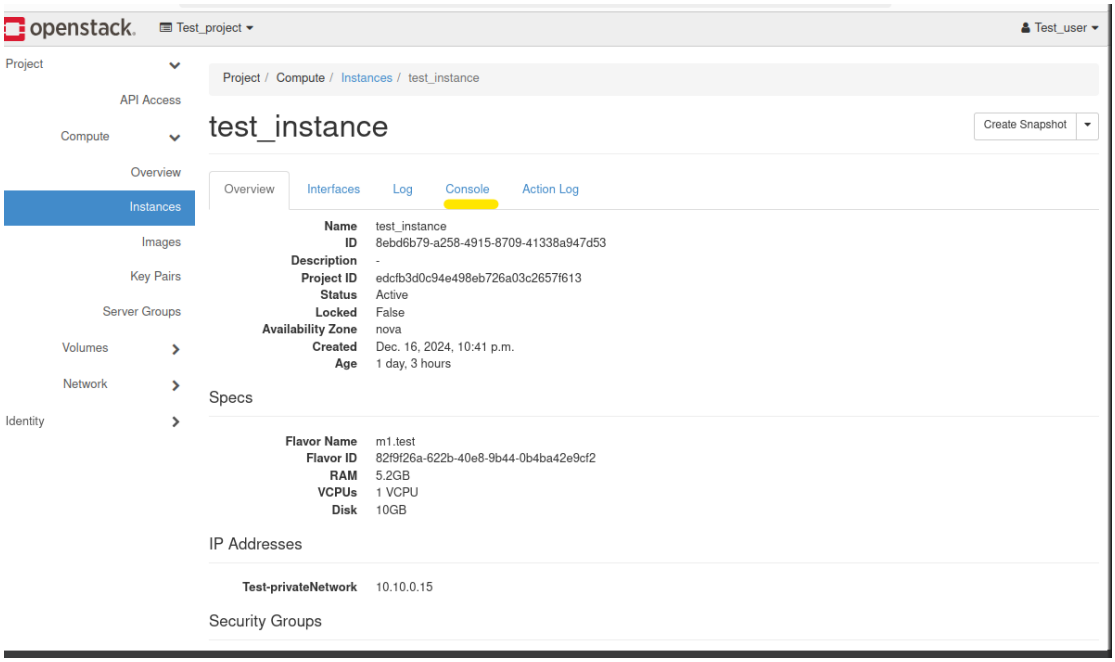


Our network topology after creating the instance.

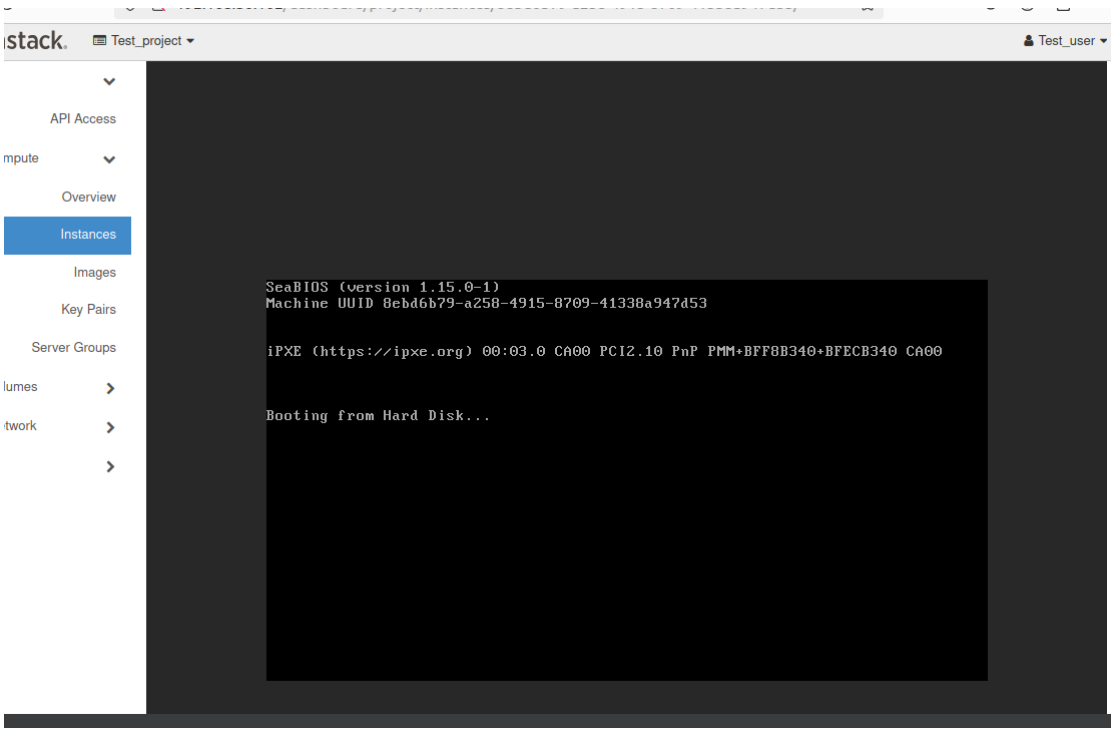
➢ Step8: Launching the instance:



If your instance is stopped, click “Start Instance” and then click the instance name.



Click on “Console” to interacte with your instance via command line.



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➢ Step10: Creating a CirrOS Instance:

Since launching an Ubuntu machine requires significant resources, we chose to launch a minimalist CirrOS machine provided by OpenStack to test instance deployment:

CirrOS is a lightweight virtual machine image specifically designed for use in cloud computing environments, including OpenStack.

- Lightweight Nature: CirrOS is an extremely lightweight and minimal Linux distribution, optimized for cloud computing environments. Its small size makes it ideal for quick deployments and agile testing.
- Fast Boot: The CirrOS image is designed for rapid booting, making it an efficient choice for scenarios where deployment speed is critical, such as automated testing and cloud experiments.

We chose the name: **cirros**.

VII. Challenges encountered

1. Network Configuration:

Network configuration is a crucial aspect when setting up a private cloud with OpenStack. Below is a detailed explanation of the potential challenges we encountered while working with Ubuntu virtual machines and OpenStack. Configuring the network in an OpenStack environment can be complex, especially if you have specific requirements for security, redundancy, or performance:

Issues with IP Address Configuration:

- Incorrect Configuration: Misconfigured IP addresses on the virtual machine or at the OpenStack level.
  Possible Solutions:
  - Check IP Configuration: Ensure that the IP addresses for the virtual machine are correctly set up. Also, verify that the OpenStack subnets are configured properly.

Routing Issues Between Private and Public Networks:

- Routing Problems: Incorrect routing between private and public networks can lead to connectivity loss.
  Possible Solutions:
  - Configure Routing: Ensure that the routing between private and public networks is set up correctly. Check the routing tables in OpenStack.

SSH Connection Issues:

- Firewall: Firewall rules may block SSH connections.
  Possible Solutions:
  - Configure Firewall Rules: Open the SSH port (typically port 22) in the virtual machine's firewall and ensure that OpenStack's security rules allow SSH traffic.

Resource Management:

Resource management is a critical aspect of setting up a private cloud, whether on a physical or virtual machine. Below are some specific challenges you may encounter, particularly when setting up a private cloud with an Ubuntu virtual machine:

RAM Allocation and Management:

- Virtual Machine Configuration: The initial configuration of the virtual machine, such as allocating RAM, virtual CPUs, and disk size, can significantly impact its resource needs and lead to instance launch issues under OpenStack.
- Overcommitment: Poor planning may result in RAM overcommitment.
  Possible Solutions:
    - Monitor RAM Usage: Use tools like `top` or `htop` to monitor RAM usage on the Ubuntu virtual machine.
    - Adjust Allocation: If necessary, adjust the RAM allocation in the virtual machine settings via OpenStack.

Storage Management:

- Storage Shortage: Storage can become an issue if virtual machines generate large amounts of data or if VM images take up too much space.
  Possible Solutions:
    - Monitor Storage Usage: Use commands to monitor storage usage on the virtual machine.
    - Resize Storage: If necessary, resize the storage volumes associated with the virtual machine via OpenStack.

Disk Performance:

- Poor Disk Performance: Virtual disk performance can be lower than physical disk performance, especially when using shared storage.
  Possible Solutions:
    - Use Fast Storage: Where possible, use faster storage solutions to enhance performance.
    - Optimize I/O: Opt for disk configurations that are optimized for specific workloads.

Conclusion

The objective of our project is to set up a private cloud using OpenStack. The first step was to provide a general overview of cloud computing, its architecture, and its various services. Subsequently, we conducted a study of different popular private cloud solutions, highlighting the virtualization techniques used in each of them.

The study enabled us to select the most suitable solution: **OpenStack**.

OpenStack has an open-source cloud computing architecture that offers a robust and scalable cloud infrastructure. Its modular architecture consists of several interconnected components, providing significant flexibility and extensibility. The key modules of OpenStack include **Nova** for virtual machine management, **Neutron** for network management, **Cinder** for block storage, **Glance** for image management, and **Keystone** for identity management. This architecture facilitates the creation of private clouds tailored to a variety of workloads. OpenStack provides capabilities such as horizontal scalability, automated resource management, high availability, and a user-friendly interface. It supports multiple hypervisors. With its advanced features, OpenStack plays a critical role in building and managing flexible and efficient cloud infrastructures.

In the area of **network configuration**, we overcame challenges such as IP address configuration and managing private and public networks. By adopting a methodical approach and following OpenStack's best practices, we were able to establish a robust network infrastructure.

Resource management was another major concern, particularly regarding RAM allocation, storage management, and disk performance optimization.

This project also allowed us to develop valuable skills in problem-solving, team collaboration, and adaptability to technological challenges. The knowledge gained on OpenStack, network configuration, and resource management has strengthened our expertise in the field of cloud infrastructures.

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